

Electromagnetism Is Relativity

Maxwell's equations never needed fixing. They were relativistic before anyone knew what that meant — which is precisely why they produced a frame-independent c and started the crisis in the first place.

WHERE WE ARE

Chapter 2.1 left you at a fork. Three things were believed in 1900 — the principle of relativity, Maxwell's equations, and the Galilean transformation — any two of them comfortable together and all three impossible at once. Branch (A) said relativity fails for optics and there is an ether frame; branch (B) said Maxwell is wrong; branch (C) said the Galilean transformation is wrong. Chapters 2.2 to 2.5 took branch (C) and rebuilt kinematics and dynamics from the wreckage.

This chapter collects the payment. Rewritten in the language of Chapter 2.4, Maxwell's four equations become **two**, of which one is not physics at all but bookkeeping. And the paradox of 2.1 §3 — c with respect to what? — evaporates, not because we answer the question but because we finally see that it was malformed. Maxwell's equations single out a *speed*. They never singled out a *frame*. Those are different things, and only after 2.2 do you have a geometry in which a speed can be frame-independent without there being anything for it to be measured relative to.

Two promises also come due. Chapter 2.4 §7.2 counted six independent components in an antisymmetric 4×4 tensor and told you they would turn out to be **E** and **B**; §2 and §3 below deliver them, by construction, with every component checked. And Chapter 1.1 §3.3 showed Newton's third law *failing* for two moving charges, left an explicit IOU marked "the field carries momentum", and told you Chapter 2.6 would pay it. Section 10 pays it, to the last factor of $\frac{3}{2}$.

TOOLS YOU'LL NEED — Chapter 0.7: divergence and curl in Cartesian components (§3.3, §4.2), the continuity equation (§6), and the two second-derivative identities $\nabla \times \nabla \phi = \mathbf{0}$ and $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ (§7), together with §7.3 on why the vector potential exists and is not unique. Chapter 1.1 §3.3: the two moving charges that break the third law — §10 here is the resolution, so it is worth rereading the setup. Chapter 1.2 §8: the field Euler–Lagrange equation, and §8.1's table of actions — §9 here supplies the third entry. Chapter 1.4 §6: Noether's theorem for fields, and its grind box on improvement terms, which §10 uses without apology. Chapter 2.1 §2 (Maxwell produces c), §4 (the wave equation is not Galilean invariant) and the fork of §4.3. Chapter 2.4 §2.1 (the boost matrix $\Lambda^\mu{}_\nu$), §4 (raising and lowering), §5 (the tensor transformation law and the quotient theorem), §6 (the invariance theorem), §7.2 (the count of six), §8.1 (the Levi-Civita symbol). Chapter 2.5 §1.3 (four-velocity), §2 (four-momentum), §6.1 (the covariant force $dp^\mu/d\tau = f^\mu$ with $f \cdot u = 0$).

► WHAT IS BEING ASSUMED, STATED ONCE

This chapter does not derive electromagnetism from nothing. Two things are taken as given.

(i) **Maxwell's four equations**, quoted in Chapter 2.1 §2 as the empirical input of that chapter, in SI units and with sources present:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \cdot \mathbf{B} &= 0, & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}$$

(ii) **Charge invariance**. The electric charge of a body is the same number in every inertial frame.

This is an experimental fact, not a theorem — and a very sharply tested one: atoms and molecules are neutral to better than one part in 10^{21} even though the electrons inside them move at wildly different speeds from the nuclei. Chapter 6.3 will make it a consequence of gauge symmetry; here it is an input, and §6 is careful about exactly how much work it does.

Everything else below is derived. What the chapter *produces* is the observation that (i) is already a Lorentz-tensor equation in disguise, that the disguise is the only thing that made it look incompatible with relativity, and that Chapter 6.3 can then run the whole construction backwards and obtain (i) from a symmetry principle.

1 · The four-current

Start with the simplest object in the theory, and build it by requirement rather than by decree.

Chapter 0.7 §6 derived, from nothing but "charge is neither created nor destroyed", the **continuity equation**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0, \tag{2.6.1}$$

where ρ is charge per unit volume and $\mathbf{J}$ is current density. Now look at (2.6.1) with the eyes of Chapter 2.4. It is a sum of four terms: one time derivative of one function, and three space derivatives of three functions. That is exactly the shape of a four-divergence $\partial_\mu V^\mu$ — if the four functions can be arranged into a four-vector.

So let us not guess. Let us demand. Suppose there exists a four-vector j^μ whose four-divergence reproduces (2.6.1), and ask what its components must be. Recall from Chapter 2.4 §8 that $\partial_\mu = \frac{\partial}{\partial x^\mu}$ carries a *lower* index, and that $x^0 = ct$, so

$$\partial_0 = \frac{\partial}{\partial x^0} = \frac{\partial}{\partial(ct)} = \frac{1}{c} \frac{\partial}{\partial t}, \quad \partial_i = \frac{\partial}{\partial x^i}. \tag{2.6.2}$$

Write the unknown four-vector as $j^\mu = (j^0, \mathbf{j})$. Then

$$\partial_\mu j^\mu = \partial_0 j^0 + \partial_i j^i = \frac{1}{c} \frac{\partial j^0}{\partial t} + \nabla \cdot \mathbf{j}. \quad (2.6.3)$$

Compare term by term with (2.6.1). The spatial part matches immediately if $\mathbf{j} = \mathbf{J}$. The temporal part matches if $j^0/c = \rho$, i.e. $j^0 = c\rho$. There is no freedom: the two expressions are equal as functions for arbitrary ρ and $\mathbf{J}$ only for that choice. Hence

$$j^\mu = (c\rho, \mathbf{J}), \quad \partial_\mu j^\mu = 0. \quad (2.6.4)$$

The factor of c is not cosmetic; it is dimensional bookkeeping forced by $x^0 = ct$, exactly as the c in $x^\mu = (ct, \mathbf{x})$ was. And notice what has happened to charge conservation. It was a four-term partial differential equation whose Lorentz behaviour was completely obscure. It is now $\partial_\mu j^\mu = 0$: a contraction of a $(0, 1)$ index against a $(1, 0)$ index, hence a *scalar* by Chapter 2.4 §5.1, hence the same statement in every frame by the invariance theorem of §6. **Charge conservation is manifestly relativistic, and it takes one line to say so.**

WHY ρ AND $\mathbf{J}$ HAD TO BE ONE OBJECT

You should find this unsurprising, and if you do not, here is the physical version. A line of static charges has $\rho \neq 0$ and $\mathbf{J} = \mathbf{0}$. Run past it and the charges are moving, so now $\mathbf{J} \neq \mathbf{0}$. Charge density and current density are the same thing seen from different frames — precisely as t and $\mathbf{x}$ were in Chapter 2.2, and as E and $\mathbf{p}$ were in Chapter 2.5. Section 6 of this chapter turns that observation into the entire explanation of magnetism.

▮ QUOTED FORWARD — WHERE THIS CURRENT WILL COME FROM

Chapter 1.4 §6 proved that a continuous symmetry of a field theory produces a conserved current satisfying $\partial_\mu j^\mu = 0$, and §6.2 of that chapter ran the machinery on a global phase rotation of a complex field. That current, once the phase is allowed to vary from point to point, is (2.6.4). We are not in a position to show that yet — it needs the gauge principle of Chapter 6.3 — so for now j^μ is constructed from measured densities. But the logical order is worth knowing in advance: conservation of charge is not an extra postulate bolted onto electromagnetism. It is a theorem about a symmetry.

▼ Grind box — why ρ alone is not a scalar, and how the γ 's work out

A common first guess is that charge density should be an invariant, since charge is. It is not, and the reason is instructive.

Take N charges q sitting at rest in a box of volume V_0 , so $\rho_0 = Nq/V_0$ in their rest frame. View them from a frame in which the box moves at speed v along x . The charge Nq is unchanged — that is the invariance assumption (ii). The volume is not: the box is contracted along x by γ , so $V = V_0/\gamma$. Hence

$$\rho = \frac{Nq}{V} = \gamma \rho_0, \quad \mathbf{J} = \rho \mathbf{v} = \gamma \rho_0 \mathbf{v}.$$

Now assemble the four-vector and compare with the four-velocity $u^\mu = \gamma(c, \mathbf{v})$ of Chapter 2.5 §1.3:

$$j^\mu = (c\gamma\rho_0, \gamma\rho_0\mathbf{v}) = \rho_0 \gamma(c, \mathbf{v}) = \rho_0 u^\mu.$$

So the four-current of a moving cloud of charge is *the invariant rest-frame density times the four-velocity*. That is manifestly a four-vector: a scalar multiplying a $(1, 0)$ tensor. Notice how the single factor of γ from length contraction is exactly the factor of γ that u^μ carries — the same γ , arriving twice for the same reason. If charge density had been invariant, j^μ would not have been a four-vector and (2.6.4) would be false.

One caution. The relation $j^\mu = \rho_0 u^\mu$ holds for a single species of charge all moving together. A copper wire has two species — a stationary lattice and a drifting electron gas — and one simply adds their four-currents. That addition is the whole of §6, and the fact that the two species carry *different* factors of γ is the whole of magnetism.

2 · The four-potential and the field tensor

Now the six components.

2.1 · Recalling the potentials

Chapter 0.7 §7.3 established the first half. Since $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ identically, writing

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{2.6.5}$$

makes $\nabla \cdot \mathbf{B} = 0$ automatic. Feed that into Faraday's law:

$$\begin{aligned} \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}) = -\nabla \times \frac{\partial \mathbf{A}}{\partial t} \\ \implies \nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) &= \mathbf{0}, \end{aligned} \tag{2.6.6}$$

the middle step being the commutation of ∂_t with the spatial derivatives inside the curl (Clairaut, as in Chapter 2.1 §2.2). A curl-free field on a simply connected region is a gradient — Chapter 0.7 §2.3 — so $\mathbf{E} + \partial_t \mathbf{A} = -\nabla \phi$ for some scalar ϕ , the minus sign being the standard convention that makes ϕ reduce to the electrostatic potential. Hence

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \tag{2.6.7}$$

Two of Maxwell's four equations have now been used up entirely: they are the *reason* the potentials exist, and once you work with ϕ and $\mathbf{A}$ they hold identically. Hold that thought; §3.4 makes it the punchline.

2.2 · Assembling A^μ

We have four functions: one scalar ϕ and three components $\mathbf{A}$. The pattern of §1 suggests trying $A^\mu = (\alpha\phi, \mathbf{A})$ for some constant α fixed by dimensions, and then *checking* that the result transforms as a four-vector rather than assuming it.

Dimensions first. In SI, $\mathbf{A}$ has units of $\text{T m} = \text{V s m}^{-1}$ and ϕ has units of V . So ϕ/c has the units of $\mathbf{A}$, and $\alpha = 1/c$ is the only choice that makes the four components commensurable — the same c , for the same reason, as in $x^\mu = (ct, \mathbf{x})$ and $j^\mu = (c\rho, \mathbf{J})$. Define

$$A^\mu \equiv \left(\frac{\phi}{c}, \mathbf{A} \right). \quad (2.6.8)$$

Is it a four-vector? Not by fiat. The check comes in §4: we will find that in the Lorenz gauge A^μ satisfies $\square A^\mu = \mu_0 j^\mu$, in which $\square$ is a scalar operator and j^μ is a four-vector by §1 — so A^μ must be one too, or the equation could not hold in every frame. Until then, treat (2.6.8) as a definition of an array of four numbers.

2.3 · The field tensor

Look at (2.6.7) again. Every entry is *one derivative of one potential component, minus another derivative of another potential component*. In four-dimensional language there is exactly one object with that shape:

$$F^{\mu\nu} \equiv \partial^\mu A^\nu - \partial^\nu A^\mu. \quad (2.6.9)$$

It is antisymmetric, $F^{\nu\mu} = -F^{\mu\nu}$, by inspection — swapping μ and ν swaps the two terms. That is not a design choice; it is forced by the construction, and the callout at the end of §2 explains why the construction itself is forced.

The upper-index derivative is $\partial^\mu = \eta^{\mu\nu} \partial_\nu$, and since $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ this flips the sign of the spatial components:

$$\partial^\mu = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\nabla \right). \quad (2.6.10)$$

That minus sign is where readers get hurt. Write it out once and keep it.

2.4 · The six components, one at a time

The three F^{0i} . Put $\mu = 0, \nu = i$ into (2.6.9):

$$\begin{aligned}
F^{0i} &= \partial^0 A^i - \partial^i A^0 \\
&= \frac{1}{c} \frac{\partial A^i}{\partial t} - (-\partial_i) \left(\frac{\phi}{c} \right) \\
&= \frac{1}{c} \left(\frac{\partial A^i}{\partial t} + \partial_i \phi \right) = -\frac{1}{c} \left(-\partial_i \phi - \frac{\partial A^i}{\partial t} \right) = -\frac{E^i}{c},
\end{aligned} \tag{2.6.11}$$

the last step being (2.6.7) read backwards. So the first row of the matrix is $-\mathbf{E}/c$, and the first column is $+\mathbf{E}/c$ by antisymmetry.

The three F^{ij} . Both indices spatial, so both derivatives pick up the minus sign from (2.6.10):

$$F^{ij} = \partial^i A^j - \partial^j A^i = -(\partial_i A^j - \partial_j A^i). \tag{2.6.12}$$

Now relate the bracket to the curl. Chapter 0.7 §4.2 wrote the curl in components; in Levi-Civita notation (Chapter 2.4 §8.1, restricted to three indices) it is $B^k = \epsilon^{klm} \partial_l A^m$. Contract with another epsilon and use the standard identity $\epsilon^{ijk} \epsilon^{klm} = \delta^{il} \delta^{jm} - \delta^{im} \delta^{jl}$:

$$\epsilon^{ijk} B^k = \epsilon^{ijk} \epsilon^{klm} \partial_l A^m = (\delta^{il} \delta^{jm} - \delta^{im} \delta^{jl}) \partial_l A^m = \partial_i A^j - \partial_j A^i. \tag{2.6.13}$$

Compare with (2.6.12):

$$F^{ij} = -\epsilon^{ijk} B^k. \tag{2.6.14}$$

Three entries, since the diagonal vanishes and the array is antisymmetric. Six in total. Assemble them:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix} \tag{2.6.15}$$

Check one entry against (2.6.14) to be sure the epsilon signs landed where they should: $F^{12} = -\epsilon^{12k} B^k = -\epsilon^{123} B^3 = -B_z \checkmark$, and $F^{13} = -\epsilon^{132} B^2 = +B_y \checkmark$ since $\epsilon^{132} = -1$.

THE PROMISE OF CHAPTER 2.4 §7.2, DELIVERED

That chapter counted: an antisymmetric rank-2 tensor in four dimensions has $\frac{1}{2} \cdot 4 \cdot 3 = 6$ independent components — too many to be a four-vector, so it must be its own kind of object. It then told you the six would turn out to be the three components of **E** and the three of **B**, and said nothing more.

Here they are. Not by analogy, not by counting coincidence: the object $\partial^\mu A^\nu - \partial^\nu A^\mu$ was built out of the potentials that Maxwell's own equations forced into existence, and its six independent entries came out as **E**/ c and **B** with no room to choose otherwise.

E and B are not two fields. They are one tensor, sliced up by an observer. Which slices you call "electric" and which "magnetic" depends on your state of motion, exactly as which part of x^μ you call "time" does. Section 5 computes the slicing rule and §6 shows you a wire where the whole of magnetism is nothing but a re-slicing.

2.5 · Both indices down, and the signs that flip

We will need $F_{\mu\nu}$ constantly, and this is exactly the step at which sign errors are manufactured. Lower with two metrics, per Chapter 2.4 §4.1:

$$F_{\mu\nu} = \eta_{\mu\alpha} \eta_{\nu\beta} F^{\alpha\beta}. \quad (2.6.16)$$

Since η is diagonal, each entry simply picks up $\eta_{\mu\mu}\eta_{\nu\nu}$ (no sum). For a $0i$ entry that is $(+1)(-1) = -1$; for an ij entry it is $(-1)(-1) = +1$. So

$$F_{0i} = -F^{0i} = +\frac{E^i}{c}, \quad F_{ij} = +F^{ij} = -\epsilon^{ijk} B^k. \quad (2.6.17)$$

⚠ ONLY THE ELECTRIC ENTRIES FLIP

Lowering both indices of F reverses the sign of the **E entries and leaves the **B** entries alone.**

One index is spatial in the first case and both are spatial in the second, so one minus sign survives in the first and two cancel in the second. If a calculation below ever comes out with **E** and **B** carrying the wrong relative sign, this is where to look first.

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix} \quad (2.6.18)$$

△ WHY THIS ISN'T OBVIOUS — ANTISYMMETRY IS NOT A CHOICE

It is tempting to read (2.6.9) as a clever guess: *somebody noticed that an antisymmetric tensor has six slots, and $\mathbf{E}$ and $\mathbf{B}$ have six components between them, so they packed them in.* That reading gets the logic backwards, and the correct order matters because Chapter 6.3 runs on it.

The chain is: **(1)** $\nabla \cdot \mathbf{B} = 0$ and Faraday's law force the existence of potentials ϕ , $\mathbf{A}$ (§2.1). **(2)** The potentials are not unique — you may add $\partial^\mu \chi$ — so the physical fields must be built from A^μ in a way that kills that freedom. **(3)** The *only* combination of one derivative and one potential that is annihilated by $A^\mu \rightarrow A^\mu + \partial^\mu \chi$ is the antisymmetric one, because $\partial^\mu \partial^\nu \chi$ is symmetric and only the antisymmetric part of $\partial^\mu A^\nu$ can avoid it. **(4)** Therefore $F^{\mu\nu}$ is antisymmetric, and therefore it has six components, and therefore they are $\mathbf{E}$ and $\mathbf{B}$.

Gauge invariance is upstream of everything. The six-component count is a *consequence*, and Chapter 2.4's promise was a prediction rather than a coincidence noticed after the fact. Notice also that the symmetric part $\partial^\mu A^\nu + \partial^\nu A^\mu$, with its ten components, is not gauge invariant and is physically meaningless here — but it is exactly the object that *does* carry physics in Chapter 3.6, where the potential is a symmetric $h_{\mu\nu}$ and the field is the linearised Riemann tensor. Same construction, different symmetry, and the difference between a spin-1 force and a spin-2 one.

▼ Grind box — every entry of $F^{\mu\nu}$ from $F = \partial A - \partial A$, by hand

Nothing here is subtle; the point is to have watched all sixteen numbers appear once, so that the matrix above is a fact rather than a memory. Write $A^\mu = (\phi/c, A_x, A_y, A_z)$ and $\partial^\mu = (\frac{1}{c}\partial_t, -\partial_x, -\partial_y, -\partial_z)$.

Diagonal. $F^{\mu\mu} = \partial^\mu A^\mu - \partial^\mu A^\mu = 0$ for each fixed μ (no sum). Four zeros.

F^{01} .

$$F^{01} = \frac{1}{c} \frac{\partial A_x}{\partial t} - \left(-\frac{\partial}{\partial x} \right) \frac{\phi}{c} = \frac{1}{c} \left(\frac{\partial A_x}{\partial t} + \frac{\partial \phi}{\partial x} \right) = -\frac{1}{c} E_x,$$

using $E_x = -\partial_x \phi - \partial_t A_x$. Identically $F^{02} = -E_y/c$ and $F^{03} = -E_z/c$.

F^{12} . Both spatial, so both derivatives carry the minus:

$$F^{12} = \left(-\frac{\partial}{\partial x} \right) A_y - \left(-\frac{\partial}{\partial y} \right) A_x = - \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) = -B_z,$$

since $B_z = \partial_x A_y - \partial_y A_x$ from Chapter 0.7 §4.2. Cycling $x \rightarrow y \rightarrow z \rightarrow x$ gives $F^{23} = -B_x$ and $F^{31} = -B_y$, and the last of these is $F^{13} = +B_y$.

Lower half. By antisymmetry, $F^{10} = +E_x/c$, $F^{21} = +B_z$, and so on. Sixteen numbers, six of them independent, exactly as (2.6.15) says.

Sanity check on units. Every entry of $F^{\mu\nu}$ must have the same dimensions, or the object is not a tensor. B is in tesla; E/c is $\text{V m}^{-1}/(\text{m s}^{-1}) = \text{V s m}^{-2} = \text{T}$ ✓. The factor of c in E/c is not decoration — without it the six components could not be six components of one thing.

3 · Four equations become two

This is the centrepiece, and nothing in it will be waved through. We are going to write down two tensor equations and expand every single component.

3.1 · The inhomogeneous half

What can we build from $F^{\mu\nu}$, one derivative, and the current? The index structure almost writes the answer. $\partial_\mu F^{\mu\nu}$ has one free upper index; so does j^ν . There is one candidate equation, and only a dimensionful constant is undetermined:

$$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu. \quad (2.6.19)$$

We now show that (2.6.19) is Gauss's law and the Ampère–Maxwell law, and that the constant is μ_0 . There are four components; we do all four.

Component $\nu = 0$. The sum over μ runs over 0, 1, 2, 3, but $F^{00} = 0$, so only the spatial terms survive:

$$\partial_\mu F^{\mu 0} = \underbrace{\partial_0 F^{00}}_{=0} + \partial_i F^{i0} = \partial_i \left(\frac{E^i}{c} \right) = \frac{1}{c} \nabla \cdot \mathbf{E}, \quad (2.6.20)$$

reading $F^{i0} = +E^i/c$ off the first column of (2.6.15). The right-hand side is $\mu_0 j^0 = \mu_0 c \rho$. Equate and multiply by c :

$$\nabla \cdot \mathbf{E} = \mu_0 c^2 \rho = \frac{\rho}{\epsilon_0}, \quad (2.6.21)$$

where the last step uses $\mu_0 \epsilon_0 = 1/c^2$ — which is not an input but Chapter 2.1's result $c = 1/\sqrt{\mu_0 \epsilon_0}$ rearranged. **That is Gauss's law**, and it also fixes the constant in (2.6.19) to be μ_0 : any other choice would give the wrong Coulomb force.

Component $\nu = j$ (spatial). Now both the time term and the space terms contribute:

$$\partial_\mu F^{\mu j} = \partial_0 F^{0j} + \partial_i F^{ij} = \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{E^j}{c} \right) + \partial_i (-\epsilon^{ijk} B^k). \quad (2.6.22)$$

Handle the second piece. Swap the first two indices of the epsilon, which costs a sign, and recognise the curl:

$$-\epsilon^{ijk}\partial_i B^k = +\epsilon^{jik}\partial_i B^k = (\nabla \times \mathbf{B})^j. \quad (2.6.23)$$

So (2.6.22) reads $-\frac{1}{c^2}\partial_t E^j + (\nabla \times \mathbf{B})^j = \mu_0 J^j$, that is,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (2.6.24)$$

That is the Ampère–Maxwell law, displacement current and all. One tensor equation, four components, two of Maxwell's four laws — and the displacement term, which Maxwell had to add by hand and which Chapter 0.7's Problem 4 showed was forced by charge conservation, is here not added at all. It is the $\mu = 0$ term of a sum that has to run over all four values of μ because the index is contracted.

3.2 · The homogeneous half

Two of Maxwell's equations are left: $\nabla \cdot \mathbf{B} = 0$ and Faraday's law. Neither contains a source, so the covariant version must have zero on the right. The object that works is the totally antisymmetrised derivative of F , written out as a cyclic sum:

$$\boxed{\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0.} \quad (2.6.25)$$

Before expanding, notice two structural facts, because they save all the work.

It is totally antisymmetric in $\lambda\mu\nu$. Swap any two of the three indices and the three terms permute into one another with an overall sign — check $\lambda \leftrightarrow \mu$: the first term becomes $\partial_\mu F_{\lambda\nu} = -\partial_\mu F_{\nu\lambda}$, and so on. So the left-hand side is an antisymmetric rank-3 object.

Therefore there are only four independent components. By the count of Chapter 2.4 §7.2 generalised, a totally antisymmetric rank-3 array in four dimensions has $\binom{4}{3} = 4$ independent entries, one for each way of choosing three distinct indices from $\{0, 1, 2, 3\}$ — namely $(1, 2, 3)$, $(0, 1, 2)$, $(0, 2, 3)$, $(0, 3, 1)$. Any triple with a repeated index vanishes identically. Four components; and we need exactly four scalar equations ($\nabla \cdot \mathbf{B} = 0$ is one, Faraday's law is three). The bookkeeping fits before we compute anything.

Triple $(\lambda\mu\nu) = (1, 2, 3)$. All three indices spatial. Using $F_{ij} = -\epsilon^{ijk} B^k$ from (2.6.17), so that $F_{23} = -B_x$, $F_{31} = -B_y$, $F_{12} = -B_z$:

$$\partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = -(\partial_x B_x + \partial_y B_y + \partial_z B_z) = -\nabla \cdot \mathbf{B} = 0. \quad (2.6.26)$$

No magnetic monopoles.

Triple $(\lambda\mu\nu) = (0, 1, 2)$. One time index. Now $F_{12} = -B_z$, $F_{20} = -F_{02} = -E_y/c$, $F_{01} = +E_x/c$, and $\partial_0 = \frac{1}{c}\partial_t$:

$$\begin{aligned}
\partial_0 F_{12} + \partial_1 F_{20} + \partial_2 F_{01} &= \frac{1}{c} \frac{\partial}{\partial t} (-B_z) + \frac{\partial}{\partial x} \left(-\frac{E_y}{c} \right) + \frac{\partial}{\partial y} \left(\frac{E_x}{c} \right) \\
&= -\frac{1}{c} \left[\frac{\partial B_z}{\partial t} + \underbrace{\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right)}_{(\nabla \times \mathbf{E})_z} \right] = 0.
\end{aligned} \tag{2.6.27}$$

Multiply by $-c$ and you have the z -component of **Faraday's law**, $(\nabla \times \mathbf{E})_z = -\partial_t B_z$. The triples $(0, 2, 3)$ and $(0, 3, 1)$ give the x - and y -components by the identical computation with the labels cycled; the grind box does them explicitly so that "by symmetry" is not doing any hidden work.

▼ Grind box — the other two Faraday components, written out

Triple $(0, 2, 3)$. The entries needed are $F_{23} = -B_x$, $F_{30} = -F_{03} = -E_z/c$, $F_{02} = +E_y/c$:

$$\begin{aligned}
\partial_0 F_{23} + \partial_2 F_{30} + \partial_3 F_{02} &= \frac{1}{c} \frac{\partial}{\partial t} (-B_x) + \frac{\partial}{\partial y} \left(-\frac{E_z}{c} \right) + \frac{\partial}{\partial z} \left(\frac{E_y}{c} \right) \\
&= -\frac{1}{c} \left[\frac{\partial B_x}{\partial t} + \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \right] = -\frac{1}{c} \left[\frac{\partial B_x}{\partial t} + (\nabla \times \mathbf{E})_x \right].
\end{aligned}$$

Triple $(0, 3, 1)$. Entries $F_{31} = -B_y$, $F_{10} = -E_x/c$, $F_{03} = +E_z/c$:

$$\begin{aligned}
\partial_0 F_{31} + \partial_3 F_{10} + \partial_1 F_{03} &= \frac{1}{c} \frac{\partial}{\partial t} (-B_y) + \frac{\partial}{\partial z} \left(-\frac{E_x}{c} \right) + \frac{\partial}{\partial x} \left(\frac{E_z}{c} \right) \\
&= -\frac{1}{c} \left[\frac{\partial B_y}{\partial t} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \right] = -\frac{1}{c} \left[\frac{\partial B_y}{\partial t} + (\nabla \times \mathbf{E})_y \right].
\end{aligned}$$

Setting each to zero gives the x - and y -components of Faraday's law. Together with (2.6.27) and (2.6.26) that is all four independent components of (2.6.25), and all four of Maxwell's homogeneous scalar equations. Nothing is left over and nothing is missing.

Why the cyclic pattern $(0, 1, 2)$, $(0, 2, 3)$, $(0, 3, 1)$ and not $(0, 1, 3)$? Because $(0, 3, 1)$ and $(0, 1, 3)$ are the same component up to a sign, and picking the cyclic representative keeps every equation's sign uniform. Choosing the other one is not an error, it is the same equation multiplied by -1 .

3.3 · The scoreboard

COVARIANT STATEMENT	COMPONENT	TRADITIONAL NAME
$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu$	$\nu = 0$	Gauss's law
	$\nu = 1, 2, 3$	Ampère–Maxwell
$\partial_{[\lambda} F_{\mu\nu]} = 0$	$(1, 2, 3)$	$\nabla \cdot \mathbf{B} = 0$
	$(0, ij)$	Faraday's law

Four vector equations, twelve-odd scalar equations in the traditional bookkeeping, compressed into two lines. And by the invariance theorem of Chapter 2.4 §6, because both lines are equations between tensors of the same type, **each holds in every inertial frame the moment it holds in one**. No chain-rule computation, no cross terms, none of the carnage of Chapter 2.1 §4.2. That is the whole purpose of the tensor apparatus and this is the first place the book collects on it.

3.4 · Two of Maxwell's equations are not physics

Now the structural point, and it deserves emphasis out of all proportion to its difficulty.

Substitute the definition (2.6.9) into (2.6.25), with all indices down so that $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$:

$$\begin{aligned}
 \partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} &= \partial_\lambda \partial_\mu A_\nu - \partial_\lambda \partial_\nu A_\mu \\
 &+ \partial_\mu \partial_\nu A_\lambda - \partial_\mu \partial_\lambda A_\nu \\
 &+ \partial_\nu \partial_\lambda A_\mu - \partial_\nu \partial_\mu A_\lambda.
 \end{aligned} \tag{2.6.28}$$

Six terms. Pair them off: the first cancels the fourth, the second cancels the fifth, the third cancels the sixth — each pair differing only by the order of two partial derivatives, which commute by Clairaut's theorem. The sum is identically zero.

HALF OF MAXWELL'S EQUATIONS ARE BOOKKEEPING

Given that $\mathbf{E}$ and $\mathbf{B}$ come from potentials, $\nabla \cdot \mathbf{B} = 0$ and Faraday's law cannot fail. They carry no information about how electromagnetism works. They are the statement that mixed partial derivatives commute, wearing a hat.

You have met both of them before, in Chapter 0.7 §7. There, $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ was six terms cancelling in pairs by Clairaut — and that identity is exactly (2.6.26). And $\nabla \times \nabla \phi = \mathbf{0}$ was the same cancellation with fewer terms — and that identity is exactly the content of Faraday's law once $\mathbf{E}$ is written in terms of ϕ and $\mathbf{A}$. The six-term cancellation in (2.6.28) is *both of Chapter 0.7's identities at once*, because in four dimensions they are one identity.

So the physical content of electromagnetism is entirely in (2.6.19): *charges and currents source the field*. The other two equations are the price of using potentials, and they are free.

► QUOTED FORWARD — THIS IS $D^2 = 0$

Chapter 0.7 §7.4 lined up gradient, curl and divergence and observed that composing two consecutive arrows gives zero, both times, and promised that in the language of differential forms both statements are the single equation $d \circ d = 0$. Chapter 3.5 supplies that language. In it, A_μ is a 1-form A , the definition $F = dA$ is (2.6.9), and (2.6.25) is $dF = ddA = 0$ — one symbol. The inhomogeneous half becomes $d\star F = \mu_0 \star j$, and the asymmetry between the two you have just noticed (one is an identity, one is a field equation) becomes the visible asymmetry between dF and $d\star F$. We flag this as a promise, not a derivation. But you now know what it will be a promise *about*.

4 · Gauge invariance

4.1 · The freedom

The potentials are not unique, and Chapter 0.7 §7.3 already said why: adding a gradient to $\mathbf{A}$ leaves $\nabla \times \mathbf{A}$ alone. In four-dimensional language the statement is cleaner and covers ϕ as well. Let $\chi(x)$ be any smooth scalar function of spacetime and transform

$$A^\mu \longrightarrow \tilde{A}^\mu = A^\mu + \partial^\mu \chi. \quad (2.6.29)$$

Compute the new field tensor directly:

$$\tilde{F}^{\mu\nu} = \partial^\mu \tilde{A}^\nu - \partial^\nu \tilde{A}^\mu = F^{\mu\nu} + \underbrace{\partial^\mu \partial^\nu \chi - \partial^\nu \partial^\mu \chi}_{=0} = F^{\mu\nu}. \quad (2.6.30)$$

Clairaut again — the third appearance in as many sections, and the last. **The fields are untouched.** Since the fields are what exert forces, no experiment can distinguish A^μ from $\tilde{A}^\mu$.

Unpack (2.6.29) into three-vector language, remembering the minus sign in $\partial^\mu = (\frac{1}{c}\partial_t, -\nabla)$:

$$\frac{\tilde{\phi}}{c} = \frac{\phi}{c} + \frac{1}{c} \frac{\partial \chi}{\partial t}, \quad \tilde{\mathbf{A}} = \mathbf{A} - \nabla \chi \quad \implies \quad \tilde{\phi} = \phi + \frac{\partial \chi}{\partial t}, \quad \tilde{\mathbf{A}} = \mathbf{A} - \nabla \chi. \quad (2.6.31)$$

(The familiar textbook form has $\mathbf{A} + \nabla \chi$ and $\phi - \partial_t \chi$; that is the same transformation with $\chi \rightarrow -\chi$, and nothing depends on which sign convention you adopt.)

YOU HAVE SEEN THIS FREEDOM BEFORE

Chapter 1.2's Problem 4 established that adding a total time derivative to a Lagrangian, $L \rightarrow L + \frac{dF}{dt}$, changes the action only by a boundary term and therefore leaves the equations of motion untouched — and Chapter 1.4 §1.2 then built the definition of a symmetry around exactly that loophole. **Gauge freedom is that freedom.** Section 9 makes the correspondence exact: the interaction term in the electromagnetic Lagrangian is $-j_\mu A^\mu$, and under (2.6.29) it changes by

$$-j_\mu \partial^\mu \chi = -\partial^\mu (j_\mu \chi) + \underbrace{\chi \partial^\mu j_\mu}_{=0} = -\partial^\mu (j_\mu \chi),$$

a pure four-divergence — the field-theory version of $\frac{dF}{dt}$, integrating to a boundary term by the divergence theorem. Note what had to be true for that to work: *charge conservation*. Gauge invariance of the action and conservation of charge are the same statement seen from two sides, which is the first hint of the structure Chapter 6.3 turns into a machine.

4.2 · Choosing a gauge, and recovering 2.1's wave equation

Freedom is a nuisance if you want to solve for something, so we spend it. Impose the **Lorenz condition**

$$\partial_\mu A^\mu = 0 \quad \iff \quad \frac{1}{c^2} \frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{A} = 0, \quad (2.6.32)$$

the equivalence following from $\partial_\mu A^\mu = \frac{1}{c} \partial_t (\phi/c) + \nabla \cdot \mathbf{A}$. Note that (2.6.32) is itself a scalar equation — a contraction — so imposing it in one frame imposes it in all. That is not true of the other common choice, the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$, which singles out a frame and is therefore useless here (though it is convenient elsewhere, and §10 uses it once, deliberately).

It is always attainable. Suppose you are handed potentials with $\partial_\mu A^\mu = f(x) \neq 0$. Gauge-transform by some χ and compute the new divergence:

$$\partial_\mu \tilde{A}^\mu = \partial_\mu A^\mu + \partial_\mu \partial^\mu \chi = f + \square \chi, \quad \square \equiv \partial_\mu \partial^\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2. \quad (2.6.33)$$

So the Lorenz condition holds for the new potentials precisely when $\square\chi = -f$. That is an inhomogeneous wave equation for χ with a known source.

► QUOTED — SOLVABILITY OF THE WAVE EQUATION

For any reasonable source f the equation $\square\chi = -f$ has solutions. This is the standard existence theorem for the wave operator (the solution is built from the retarded Green's function, $\chi(\mathbf{x}, t) = \frac{1}{4\pi} \int f(\mathbf{x}', t - |\mathbf{x} - \mathbf{x}'|/c) / |\mathbf{x} - \mathbf{x}'| d^3x'$), and we quote it rather than prove it — the proof is a chapter of analysis, not of physics, and Chapter 5.6 constructs the Green's functions properly when it needs the propagator. What matters here is only that a solution exists, so the Lorenz gauge is always available.

Note also that it is not *unique*: any χ_0 with $\square\chi_0 = 0$ may be added, which is a leftover freedom called the residual gauge. That leftover is precisely why the photon has two polarisation states and not four, and Chapter 5.6 spends real effort on it.

Now the payoff. Substitute the definition of F into the field equation (2.6.19) and impose (2.6.32):

$$\begin{aligned} \mu_0 j^\nu &= \partial_\mu F^{\mu\nu} = \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= (\partial_\mu \partial^\mu) A^\nu - \partial^\nu \underbrace{(\partial_\mu A^\mu)}_{=0} = \square A^\nu. \end{aligned} \quad (2.6.34)$$

The middle step moved ∂_μ past ∂^ν , legitimate because partial derivatives commute. So all of electromagnetism, in the Lorenz gauge, is

$$\boxed{\square A^\mu = \mu_0 j^\mu.} \quad (2.6.35)$$

Four uncoupled wave equations, one per component, each with its own source. In vacuum ($j^\mu = 0$) this is $\square A^\mu = 0$, that is,

$$\frac{1}{c^2} \frac{\partial^2 A^\mu}{\partial t^2} = \nabla^2 A^\mu, \quad (2.6.36)$$

which is **the wave equation of Chapter 2.1 §2** — the one derived there by taking the curl of Faraday's law and grinding through $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$. Same equation, same c , obtained here in three lines because the bookkeeping was done first.

THE CRISIS OF 2.1, DISSOLVED

Chapter 2.1 §3 asked the question that broke nineteenth-century physics: the wave equation contains a speed c , every other wave equation's speed is measured relative to a medium, so what is c measured relative to?

Look at (2.6.35) and see why the question has no answer. The operator $\square = \partial_\mu \partial^\mu$ is a *scalar*: two indices contracted, so by Chapter 2.4 §5.1 it takes the same form in every inertial frame, and c enters it only through $\eta_{\mu\nu}$ — through the **metric of spacetime itself**. The constant c in Maxwell's equations is a property of the geometry in which the fields live, not of any substance they live in. Asking what it is measured relative to is like asking what the number π is measured relative to.

And now branch (A) of 2.1's fork is not merely unsupported by Michelson and Morley — it is structurally unavailable. Maxwell's equations do pick out a preferred *speed*. They never picked out a preferred *frame*, and only the assumption that $t' = t$ made those two look like the same claim.

Electromagnetism was never the theory that needed fixing. It was relativistic from birth, and it was mechanics — with its Galilean addition of velocities — that had to give way. Chapter 2.5 has already paid that bill.

WHERE GAUGE FREEDOM IS GOING

In this chapter gauge invariance is a convenience: it lets you choose (2.6.32) and turn a coupled mess into four wave equations. That is a wild understatement of its importance.

Chapter 6.3 reverses the logic. Instead of noticing that electromagnetism happens to have this redundancy, it *demand*s that the phase of a charged quantum field be adjustable independently at every point of spacetime — a symmetry with a function's worth of parameters — and finds that the demand cannot be met unless a vector field A^μ exists, transforming exactly as (2.6.29), and coupling exactly as $-j_\mu A^\mu$. The entire content of this chapter comes out as a *consequence*. Repeat the trick with a non-commuting symmetry group and out come the weak and strong interactions (Chapter 6.4). Gauge freedom stops being a convenience and becomes the generating principle of every known force except gravity — and gravity turns out to be the same trick applied to the Lorentz group.

△ WHY THIS ISN'T OBVIOUS — THE POTENTIALS KNOW MORE THAN THE FIELDS

The natural reading of §4.1 is that A^μ is a convenient fiction and $F^{\mu\nu}$ is the real thing: the fields are gauge invariant, the potentials are not, so only the fields can be physical. That reading is *almost* right and the exception is one of the most instructive facts in physics.

Chapter 0.7 §2.4 built a vector field on the punctured plane that is curl-free *everywhere* and yet has circulation 2π around every loop enclosing the puncture. Its potential exists locally and is the polar angle, which is multivalued. Now read that as electromagnetism. Take a long solenoid with $\mathbf{B} \neq \mathbf{0}$ inside and $\mathbf{B} = \mathbf{0}$ everywhere outside. Outside, $\mathbf{A}$ is curl-free — but it cannot be set to zero, because $\oint \mathbf{A} \cdot d\mathbf{r}$ around a loop enclosing the solenoid equals the enclosed flux, by Stokes' theorem, and that is not zero. The region outside the solenoid is not simply connected, and Chapter 0.7's counterexample is exactly this situation.

Classically nothing follows: no charge outside ever feels a force. Quantum mechanically something does. Chapter 5.8 shows that a charged particle's amplitude picks up a phase $\exp\left(\frac{iq}{\hbar} \int \mathbf{A} \cdot d\mathbf{r}\right)$ along its path, so two paths passing on opposite sides of the solenoid differ in phase by $q\Phi/\hbar$ and the interference pattern **shifts** — even though the particle never enters a region where the field is nonzero. This is the **Aharonov-Bohm effect**, it was measured in 1960, and it settles the question: the potential carries information that the field does not, and the information is topological. What is physical is neither A^μ (gauge dependent) nor $F^{\mu\nu}$ (too little) but the gauge-invariant loop integral $\oint A_\mu dx^\mu$ — the *holonomy*. Chapter 6.3 builds the whole of gauge theory on that object, and Chapter 6.5 finds that in the strong interaction it is essentially all there is.

5 · $\mathbf{E}$ and $\mathbf{B}$ mix

If $\mathbf{E}$ and $\mathbf{B}$ really are the components of one tensor, then a boost must shuffle them into each other, and Chapter 2.4 tells us exactly how. For a $(2, 0)$ tensor the law is

$$F'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta F^{\alpha\beta}, \quad (2.6.37)$$

which in matrix notation, with F the array of (2.6.15), is simply $F' = \Lambda F \Lambda^\top$ — one Λ acting on rows, one on columns. Take the standard boost of Chapter 2.4 §2.1, S' moving at $+v$ along x :

$$\Lambda^\mu_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \beta = \frac{v}{c}. \quad (2.6.38)$$

Carrying out the two matrix multiplications (grind box) and reading the primed fields off the result gives the transformation rules in full.

HOW $\mathbf{E}$ AND $\mathbf{B}$ TRANSFORM UNDER A BOOST ALONG x

Splitting each field into the component along the boost ($\parallel$) and the two perpendicular to it ($\perp$):

$$\begin{aligned} E'_{\parallel} &= E_{\parallel}, & B'_{\parallel} &= B_{\parallel}, \\ E'_y &= \gamma(E_y - vB_z), & B'_y &= \gamma\left(B_y + \frac{v}{c^2}E_z\right), \\ E'_z &= \gamma(E_z + vB_y), & B'_z &= \gamma\left(B_z - \frac{v}{c^2}E_y\right). \end{aligned}$$

Or, compactly, with $\mathbf{v}$ the boost velocity:

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})_{\perp}, \quad \mathbf{B}'_{\perp} = \gamma\left(\mathbf{B} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}\right)_{\perp}.$$

Three things are worth noticing immediately.

The components along the boost do not change. That is the opposite of what happens to a four-vector, whose *parallel* component is the one that mixes with time. The reason is visible in (2.6.15): E_x sits in the 01 slot, and both of its indices are in the block the boost acts on, so it acquires two factors of Λ that undo each other. B_x sits in the 23 slot, which the boost does not touch at all.

The combination $\mathbf{E} + \mathbf{v} \times \mathbf{B}$ is already familiar. It is the Lorentz force per unit charge. Section 8 shows that is not a coincidence.

If $\mathbf{B} = \mathbf{0}$ in one frame, $\mathbf{B}' \neq \mathbf{0}$ in another — unless $\mathbf{E}$ happens to be parallel to $\mathbf{v}$. A pure electric field is not a frame-independent notion. *Magnetism is what an electric field looks like from a moving frame*, and §6 makes that quantitative in the one case where you can check it against a laboratory measurement.

▼ Grind box — the boost of $F^{\mu\nu}$, entry by entry

The transformation is $F' = \Lambda F \Lambda^T$. Since Λ is symmetric here, $\Lambda^T = \Lambda$, and it acts nontrivially only on the 0-1 block. Write Λ as the 2×2 block $\begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}$ on indices (0, 1) and the identity on (2, 3).

Entries with both indices in $\{0, 1\}$. Only F^{01} is nonzero there. The 2×2 block of F is $\begin{pmatrix} 0 & -E_x/c \\ E_x/c & 0 \end{pmatrix}$, i.e. $(-E_x/c)$ times the antisymmetric symbol. For any 2×2 matrix M and the antisymmetric ε , $M\varepsilon M^T = \det(M)\varepsilon$ — a two-dimensional special case of the determinant identity in Chapter 2.4 §8.1. Here $\det = \gamma^2 - \gamma^2\beta^2 = \gamma^2(1 - \beta^2) = 1$. Hence

$$F'^{01} = F^{01} \quad \implies \quad E'_x = E_x.$$

Entries with both indices in $\{2, 3\}$. Λ is the identity there, so $F'^{23} = F^{23}$, i.e. $B'_x = B_x$.

Mixed entries. These carry one factor of the block and one identity. For F'^{02} :

$$F'^{02} = \Lambda^0_\alpha \Lambda^2_\beta F^{\alpha\beta} = \Lambda^0_\alpha F^{\alpha 2} = \gamma F^{02} - \gamma\beta F^{12}.$$

Substitute $F^{02} = -E_y/c$ and $F^{12} = -B_z$:

$$-\frac{E'_y}{c} = -\gamma \frac{E_y}{c} + \gamma\beta B_z \quad \Longrightarrow \quad E'_y = \gamma(E_y - \beta c B_z) = \gamma(E_y - v B_z).$$

For F'^{03} , with $F^{03} = -E_z/c$ and $F^{13} = +B_y$:

$$-\frac{E'_z}{c} = \gamma F^{03} - \gamma\beta F^{13} = -\gamma \frac{E_z}{c} - \gamma\beta B_y \quad \Longrightarrow \quad E'_z = \gamma(E_z + v B_y).$$

For F'^{12} , now the first index sits in the block as a row:

$$F'^{12} = \Lambda^1_\alpha F^{\alpha 2} = -\gamma\beta F^{02} + \gamma F^{12} = \gamma\beta \frac{E_y}{c} - \gamma B_z,$$

and since $F'^{12} = -B'_z$, $B'_z = \gamma(B_z - v E_y/c^2)$. Identically, $F'^{13} = -\gamma\beta F^{03} + \gamma F^{13}$ gives $B'_y = \gamma(B_y + v E_z/c^2)$.

Six components, six rules, no others. Every one of them was checked symbolically against $\Lambda F \Lambda^\top$ before this box was written.

A consistency check worth doing. Apply the rules twice, once with v and once with $-v$, and you must get back what you started with. For E_y : $\gamma(E'_y + v B'_z) = \gamma^2[(E_y - v B_z) + v(B_z - v E_y/c^2)] = \gamma^2 E_y(1 - v^2/c^2) = E_y \checkmark$. The $\gamma^2(1 - \beta^2) = 1$ that makes it work is the same identity that made $\det \Lambda = 1$ above.

⚠ "E AND B ARE THE SAME THING" — WHAT THAT DOES AND DOES NOT MEAN

It does *not* mean you can turn an electric field into a magnetic field by running. The invariants of §7 forbid it in general: if $\mathbf{E} \cdot \mathbf{B} \neq 0$ in one frame it is nonzero in every frame, so a field configuration with both fields present and non-orthogonal has no frame where either vanishes.

What it means is precise and weaker: **E and B are components of a single object, and the split between them is observer-dependent in exactly the way the split of a four-vector into time and space parts is.** Two observers disagree about how much of $F^{\mu\nu}$ is electric in the same sense that they disagree about how much of p^μ is energy — not because either is confused, but because "electric" names a slice and they are slicing differently. What they agree about is $F^{\mu\nu}$, and the two scalars §7 builds from it.

6 · The wire — magnetism as electrostatics in disguise

This is the argument that makes the chapter's title literal, and it can be checked against a current balance on a laboratory bench.

6.1 · The setup, in the lab

Take a long straight wire along the x -axis. Model it, as Chapter 1.1's grind boxes model everything, with the crudest structure that has the right physics: a rigid lattice of positive ions at rest, and a gas of conduction electrons drifting through it.

- **Lattice:** at rest in the lab, linear charge density $+\lambda_0$.
- **Electrons:** drifting with velocity $-v_d\hat{\mathbf{x}}$, linear charge density $-\lambda_0$ as measured in the lab.

The two densities are equal and opposite *in the lab*, so the wire is electrically neutral there — which is an experimental fact about wires, not an assumption: a current-carrying wire does not attract a stationary pith ball. The conventional current is

$$I = (-\lambda_0) \times (-v_d) = +\lambda_0 v_d, \quad (2.6.39)$$

flowing in the $+\hat{\mathbf{x}}$ direction, because negative charge moving one way is positive current the other.

Now put a test charge $q > 0$ at perpendicular distance r from the wire, at position $r\hat{\mathbf{y}}$, moving parallel to the wire with velocity $u\hat{\mathbf{x}}$.

In the lab the wire is neutral, so $\mathbf{E} = \mathbf{0}$ and there is no electric force whatever. The magnetic field of a long straight wire at distance r is $\mathbf{B} = \frac{\mu_0 I}{2\pi r} \hat{\mathbf{x}} \times \hat{\mathbf{r}}$; here $\hat{\mathbf{r}} = \hat{\mathbf{y}}$ and $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$, so

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r} \hat{\mathbf{z}}, \quad \mathbf{F}_{\text{lab}} = q u \hat{\mathbf{x}} \times \mathbf{B} = -\frac{\mu_0 q u \lambda_0 v_d}{2\pi r} \hat{\mathbf{y}}. \quad (2.6.40)$$

Attractive — the force points from the charge toward the wire, since $\hat{\mathbf{x}} \times \hat{\mathbf{z}} = -\hat{\mathbf{y}}$. **The force is purely magnetic and there is no ambiguity about that.**

6.2 · The same situation, from the test charge's frame

Now board the test charge. Transform to the frame S' moving at $+u\hat{\mathbf{x}}$. In S' the test charge is at rest, so *it cannot feel a magnetic force at all* — the Lorentz force on a stationary charge has no magnetic term, whatever $\mathbf{B}'$ may be. And yet the charge is certainly still accelerating toward the wire; whether it hits the wire is not a matter of opinion. Some other force must be doing the work, and there is only one candidate.

Compute the two charge densities in S' . This is the crux, so we go slowly.

The lattice. At rest in the lab with density λ_0 ; therefore its *proper* density — the density in its own rest frame — is λ_0 . In S' it moves at speed u , so the lattice is length-contracted by γ_u and the same charge occupies less length:

$$\lambda'_+ = \gamma_u \lambda_0, \quad \gamma_u = (1 - u^2/c^2)^{-1/2}. \quad (2.6.41)$$

The electrons. These were already moving in the lab, so we must first back out their proper density. In the lab they move at v_d and have density $-\lambda_0$; running (2.6.41)'s logic backwards, their proper density is $-\lambda_0/\gamma_d$ with $\gamma_d = (1 - v_d^2/c^2)^{-1/2}$. In S' their velocity is, by the velocity-addition law of Chapter 2.2 §5,

$$v'_- = \frac{-v_d - u}{1 + v_d u/c^2}, \quad (2.6.42)$$

and the corresponding Lorentz factor is (grind box)

$$\gamma'_- = \gamma_u \gamma_d \left(1 + \frac{u v_d}{c^2}\right). \quad (2.6.43)$$

Multiply the proper density by that factor:

$$\lambda'_- = -\frac{\lambda_0}{\gamma_d} \cdot \gamma'_- = -\gamma_u \lambda_0 \left(1 + \frac{u v_d}{c^2}\right). \quad (2.6.44)$$

The two densities have been contracted by different factors. Both picked up γ_u ; the electrons picked up an extra $(1 + u v_d/c^2)$ because they were already moving. Add them:

$$\boxed{\lambda' = \lambda'_+ + \lambda'_- = \gamma_u \lambda_0 \left[1 - \left(1 + \frac{u v_d}{c^2}\right)\right] = -\gamma_u \lambda_0 \frac{u v_d}{c^2}.} \quad (2.6.45)$$

The wire is not neutral in this frame. It carries a net negative charge per unit length, and since the test charge is positive, it is attracted — which is the direction we already know the force to have. To leading order in $u v_d/c^2$, $\gamma_u \rightarrow 1$ and $\lambda' \approx -\lambda_0 u v_d/c^2$.

6.3 · The force, and the comparison

The field of an infinite line charge λ' at perpendicular distance r' is $E' = \lambda'/(2\pi\epsilon_0 r')$, radial. Distances perpendicular to the boost are unchanged, so $r' = r$. The force on the (stationary, in S') test charge is purely electric:

$$\mathbf{F}' = q\mathbf{E}' = \frac{q\lambda'}{2\pi\epsilon_0 r} \hat{\mathbf{y}} = -\frac{\gamma_u q \lambda_0 u v_d}{2\pi\epsilon_0 c^2 r} \hat{\mathbf{y}} = -\frac{\mu_0 \gamma_u q u \lambda_0 v_d}{2\pi r} \hat{\mathbf{y}}, \quad (2.6.46)$$

using $1/(\epsilon_0 c^2) = \mu_0$ in the last step. Compare with the lab result (2.6.40):

$$\boxed{\mathbf{F}' = \gamma_u \mathbf{F}_{\text{lab}}.} \quad (2.6.47)$$

And γ_u is *exactly* the factor by which a transverse force is expected to change between the two frames. Here is why, in three lines. Momentum transverse to the boost is unchanged, $p'_y = p_y$, because p^μ is a four-vector and the boost matrix of Chapter 2.4 §2.1 acts as the identity on the y and z rows. For a particle instantaneously at rest in S' we have $dx' = 0$, so the time transformation gives $dt = \gamma_u(dt' + u dx'/c^2) = \gamma_u dt'$. Hence

$$F_y = \frac{dp_y}{dt} = \frac{dp'_y}{\gamma_u dt'} = \frac{F'_y}{\gamma_u}, \quad (2.6.48)$$

which is (2.6.47) rearranged. The two calculations agree identically, not approximately.

WHAT JUST HAPPENED

One observer says: the wire is neutral, the charge is moving, there is a magnetic field, and the force is $qu \times \mathbf{B}$.

The other says: there is no magnetic force here because nothing is moving; the wire carries a net negative charge, and this is Coulomb attraction.

They are both right, they compute the same trajectory, and neither description is more fundamental. **What one observer calls magnetism, another calls electrostatics acting on a wire that is not neutral.** Magnetism is a relativistic correction to Coulomb's law — the v^2/c^2 piece — and the only reason it is not a fantastically small effect is that the zeroth-order term has been cancelled to fantastic precision by the neutrality of matter.

6.4 · How small is v_d/c , and why you can still pick up a nail

Put numbers in, because they are startling. Copper has one conduction electron per atom, density 8960 kg m^{-3} and molar mass 63.55 g mol^{-1} , so the number density of mobile electrons is

$$n = \frac{8960}{63.55 \times 10^{-3}} \times 6.022 \times 10^{23} = 8.49 \times 10^{28} \text{ m}^{-3}. \quad (2.6.49)$$

In a wire of cross-section 1 mm^2 carrying 1 A , the drift speed follows from $I = neAv_d$:

$$v_d = \frac{I}{neA} = \frac{1}{(8.49 \times 10^{28})(1.602 \times 10^{-19})(10^{-6})} = 7.35 \times 10^{-5} \text{ m s}^{-1}, \quad (2.6.50)$$

that is, 0.074 mm s^{-1} — slower than a growing fingernail. And

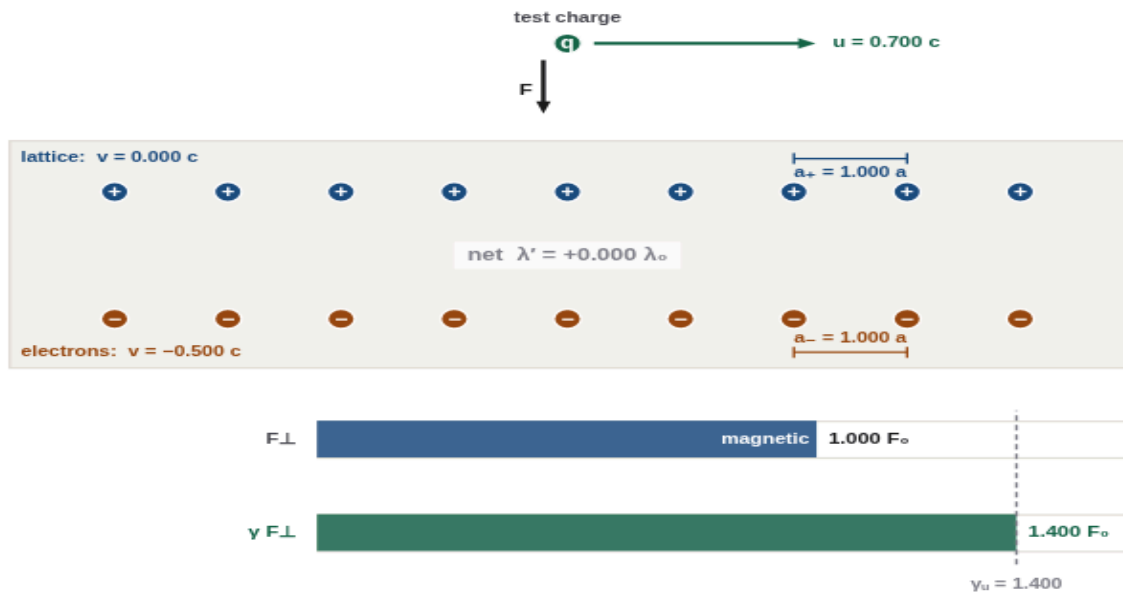
$$\frac{v_d}{c} = 2.45 \times 10^{-13}. \quad (2.6.51)$$

So magnetism is an effect of relative order 10^{-13} , and yet electromagnets lift cars. The resolution is in (2.6.45): the surviving effect is not $\lambda_0(v_d/c)$ but the *difference* between two enormous numbers, and the enormous number is λ_0

itself. In that same wire,

$$\lambda_0 = neA = 1.36 \times 10^4 \text{ C m}^{-1}, \quad (2.6.52)$$

thirteen thousand coulombs of mobile charge per metre, exactly cancelled by thirteen thousand coulombs of lattice charge. A tiny fractional imbalance in a colossal cancellation is still a substantial charge. Worked example 1 does the arithmetic all the way to a force.



boost toward q 's frame $w = 0.000\ c$

drift v_d/c (exaggerated) 0.500

lab frame

test-charge frame

maximise the drift

frame speed $w = 0.000\ c$, $\gamma_w = 1.000$ | test charge in this frame: $u = 0.700\ c$

$\lambda_+/\lambda_0 = +1.000$ $\lambda_-/\lambda_0 = -1.000 \rightarrow \text{net } 0.000$ (formula $-\gamma_w w \cdot v_d/c^2 = 0.000$)

spacings: $a_+ = 1.000\ a$, $a_- = 1.000\ a$, ratio $a_-/a_+ = 1.000$ $(1+w v_d/c^2)^{-1} = 1.000$

$F_{\text{electric}} = 0.000\ F_0$, $F_{\text{magnetic}} = 1.000\ F_0$, total = $1.000\ F_0$ | INVARIANT $\gamma \cdot F_{\perp} = 1.400\ F_0$ ($\gamma_u = 1.400$)

a real copper wire ($1\ \text{mm}^2$, $1\ \text{A}$): $v_d/c = 2.45\text{e-}13$, $\lambda_0 = 1.360\text{e+}4\ \text{C/m}$, net $\lambda' = -1.113\text{e-}12\ \text{C/m}$ at $u = 10^5\ \text{m/s} \Rightarrow F = 3.204\text{e-}19\ \text{N}$

One wire, two frames. The test charge q moves at a fixed $u = 0.7c$ in the lab; the slider carries you continuously from the lab frame to q 's rest frame. *The drift speed is exaggerated by about twelve orders of magnitude* so that the contraction is visible at all — in a real copper wire $v_d/c \approx 2.5 \times 10^{-13}$, as (2.6.51) computes and the third button reports. Everything drawn and every number displayed is evaluated live from (2.6.41)–(2.6.45) and the field transformation rules of §5; nothing is a stored result. **(i)** Watch the two rows of charges. Both compress as you boost, but *by different factors*: the lattice by γ_w , the electrons by $\gamma_w(1 + wv_d/c^2)$. The spacing readout shows the ratio, which is $(1 + wv_d/c^2)^{-1}$ and has nothing to do with how the picture was drawn. **(ii)** The net line charge appears as soon as you leave the lab frame, growing from zero to $-\gamma_u \lambda_0 wv_d/c^2$. **(iii)** The force budget bar splits the transverse force into its electric (orange) and magnetic (blue) parts. In the lab it is all magnetic; in q 's frame it is all electric. The *total* is not constant, and should not be: it grows by γ , because forces are not invariants. What *is* constant is the lower bar, $\gamma_q F_{\perp}$ — the transverse component of the four-force of Chapter 2.5 §6.1, which is a component of a four-vector and therefore untouched by a boost along x . That bar does not move by a pixel as you drag, and it is the honest statement of "the physics is the same".

▼ Grind box — the composition identity $\gamma'_- = \gamma_u \gamma_d (1 + uv_d/c^2)$

Let a particle move at velocity w along x in frame S , and let S' move at $+u$. The velocity in S' is $w' = (w - u)/(1 - uw/c^2)$. Compute $1 - w'^2/c^2$:

$$1 - \frac{w'^2}{c^2} = 1 - \frac{(w - u)^2}{c^2 \left(1 - \frac{uw}{c^2}\right)^2} = \frac{c^2 \left(1 - \frac{uw}{c^2}\right)^2 - (w - u)^2}{c^2 \left(1 - \frac{uw}{c^2}\right)^2}.$$

Expand the numerator, writing $a = u/c$, $b = w/c$ and factoring out c^2 :

$$(1 - ab)^2 - (b - a)^2 = 1 - 2ab + a^2b^2 - b^2 + 2ab - a^2 = (1 - a^2)(1 - b^2).$$

The cross terms cancel exactly — that is the only step with any content. Hence

$$1 - \frac{w'^2}{c^2} = \frac{(1 - u^2/c^2)(1 - w^2/c^2)}{(1 - uw/c^2)^2} \implies \gamma_{w'} = \gamma_u \gamma_w \left(1 - \frac{uw}{c^2}\right).$$

For the electrons, $w = -v_d$, and the minus sign in the bracket becomes a plus: $\gamma'_- = \gamma_u \gamma_d (1 + uv_d/c^2)$, which is (2.6.43). ✓

Bonus. The same identity, read in reverse, is the statement that $\gamma_{w'} = \gamma_u \gamma_w (1 - uw/c^2)$ is the 0-component of the boosted four-velocity — i.e. it is $u'^0 = \Lambda^0_\mu u^\mu$ written out. Nothing new was needed; velocity addition is the transformation of u^μ , as Chapter 2.5 remarked.

And the same identity again. Notice that the spacings drawn in the figure obey $a'_-/a'_+ = (1 + uv_d/c^2)^{-1}$, which is this identity divided by itself. The two rows of charges compress by different amounts because they enter the boost with different γ 's, and for no other reason.

△ WHY THIS ISN'T OBVIOUS — WHAT THE WIRE ARGUMENT DOES NOT PROVE

"Magnetism is just relativity applied to electrostatics" is a true and beautiful slogan and it is routinely overstated. Precisely what was used above:

1. **Coulomb's law**, for the field of a static line charge. Input.
2. **Charge invariance**. Without it, $\lambda'_+ = \gamma_u \lambda_0$ is false and the whole calculation collapses. This is an experimental fact — the ▀ callout at the head of the chapter — not a consequence of relativity.
3. **The field concept** — that the force is mediated by something local, so that "the field at the test charge" is a meaningful quantity to transform. A pure action-at-a-distance theory would need a different argument.
4. **The Lorentz force law** in the lab, to have something to compare against.

What relativity supplies is the *relation* between the two descriptions, and it supplies it with no freedom at all. Given Coulomb, charge invariance and the field concept, the magnetic force is not optional and its magnitude is fixed. That is a very strong statement — it means the constant μ_0 is not independent of ϵ_0 , which is Chapter 2.1's $c = 1/\sqrt{\mu_0\epsilon_0}$ arriving from the other direction. But it is not the same as "magnetism can be derived from electrostatics alone", which is false.

7 · The invariants

If **E** and **B** separately are observer-dependent, what do all observers agree about? By Chapter 2.4 §5.1 the answer is: whatever you can build by contracting every index. From a single antisymmetric rank-2 tensor there are exactly two such quantities.

7.1 · The first invariant

Contract F with itself:

$$F_{\mu\nu}F^{\mu\nu} = \sum_{\mu,\nu} F_{\mu\nu}F^{\mu\nu}. \quad (2.6.53)$$

Split the double sum by index type. The diagonal contributes nothing. The $0i$ and $i0$ entries contribute twice over (once each way), and by (2.6.17) $F_{0i}F^{0i} = (E^i/c)(-E^i/c) = -(E^i)^2/c^2$, so those terms give

$$2 \sum_i F_{0i}F^{0i} = -\frac{2}{c^2}(E_x^2 + E_y^2 + E_z^2) = -\frac{2E^2}{c^2}. \quad (2.6.54)$$

For the purely spatial entries, $F_{ij} = F^{ij} = -\epsilon^{ijk}B^k$, so

$$\sum_{i,j} F_{ij} F^{ij} = \sum_{i,j} \epsilon^{ijk} \epsilon^{ijl} B^k B^l = 2 \delta^{kl} B^k B^l = 2 B^2, \quad (2.6.55)$$

using the contraction identity $\epsilon^{ijk} \epsilon^{ijl} = 2 \delta^{kl}$ (sum on i, j), which is the previous identity $\epsilon^{ijk} \epsilon^{klm} = \delta^{il} \delta^{jm} - \delta^{im} \delta^{jl}$ with one more index contracted. Adding:

$$\boxed{F_{\mu\nu} F^{\mu\nu} = 2 \left(B^2 - \frac{E^2}{c^2} \right).} \quad (2.6.56)$$

7.2 · The second invariant

The other contraction available uses the Levi-Civita symbol of Chapter 2.4 §8.1:

$$\epsilon_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}. \quad (2.6.57)$$

Total antisymmetry means the only surviving terms are those in which $\{\mu\nu\}$ and $\{\rho\sigma\}$ are complementary pairs. There are three such pair-splittings — $\{01|23\}, \{02|13\}, \{03|12\}$ — and each accounts for eight of the 24 nonzero permutations: two orderings inside the first pair, two inside the second, and two choices of which pair comes first. Flipping any of those three costs a sign in ϵ and a sign in one factor of F , so all eight terms are *equal*. Hence

$$\begin{aligned} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma} &= 8 \left[\epsilon_{0123} F^{01} F^{23} + \epsilon_{0213} F^{02} F^{13} + \epsilon_{0312} F^{03} F^{12} \right] \\ &= 8 \left[\left(-\frac{E_x}{c} \right) (-B_x) - \left(-\frac{E_y}{c} \right) (B_y) + \left(-\frac{E_z}{c} \right) (-B_z) \right], \end{aligned} \quad (2.6.58)$$

using $\epsilon_{0123} = +1$, $\epsilon_{0213} = -1$ (one transposition) and $\epsilon_{0312} = +1$ (two transpositions), together with the entries of (2.6.15). All three terms come out positive:

$$\boxed{\epsilon_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma} = \frac{8}{c} \mathbf{E} \cdot \mathbf{B}.} \quad (2.6.59)$$

► IT IS A **PSEUDOSCALAR**, PER CHAPTER 2.4 §8.1

That chapter proved $\epsilon'_{\mu\nu\rho\sigma} = (\det \Lambda)^{-1} \epsilon_{\mu\nu\rho\sigma}$ and noted that $\det \Lambda = \pm 1$ for Lorentz transformations. So (2.6.59) is genuinely invariant under boosts and rotations ($\det \Lambda = +1$) but changes sign under a spatial reflection. It is a **pseudoscalar**. That is not a defect; it is a fact about $\mathbf{E} \cdot \mathbf{B}$, which is the dot product of a vector with a pseudovector. Chapter 2.4 also flagged, and we re-flag, that for general coordinate changes ϵ is a tensor *density* and needs a factor of $\sqrt{-g}$; within Part II, where $\det \Lambda = \pm 1$, that never bites. A term proportional to (2.6.59) may legally be added to the Lagrangian of §9 and is the notorious θ -term; Chapter 6.5 explains why the strong interaction's version of it is measured to be astonishingly close to zero.

7.3 · What the invariants forbid

Two numbers, agreed on by everyone. They are strong constraints, and it is worth being exact about what each rules out.

A plane light wave stays a plane light wave. Chapter 2.1 §2 found that in an electromagnetic wave $\mathbf{E} \perp \mathbf{B}$ and $E = cB$. Then both invariants vanish:

$$B^2 - \frac{E^2}{c^2} = B^2 - B^2 = 0, \quad \mathbf{E} \cdot \mathbf{B} = 0. \quad (2.6.60)$$

Both are zero in *every* frame, so $E' = cB'$ and $\mathbf{E}' \perp \mathbf{B}'$ for every observer, no matter how fast. You may Doppler-shift a light wave to any frequency you like and change its amplitude by any factor — Chapter 2.5 §7.2 computed exactly how — but you can never make it into something other than a light wave. In particular **you cannot bring light to rest**: a stationary field configuration would have to have $\mathbf{E}$ and $\mathbf{B}$ independent, and there is no frame in which the ratio E/cB is anything but 1.

When can you kill the magnetic field? Suppose there exists a frame with $\mathbf{B}' = \mathbf{0}$. Then in that frame the invariants read $-E'^2/c^2$ and 0. Since invariants are invariants, that requires

$$B^2 - \frac{E^2}{c^2} < 0 \quad (\text{i.e. } E > cB), \quad \text{and} \quad \mathbf{E} \cdot \mathbf{B} = 0, \quad (2.6.61)$$

in *every* frame, in particular in yours. So the condition is checkable without leaving home: the fields must be perpendicular and the electric one must dominate. (These conditions are also sufficient — Worked Example 2 constructs the frame explicitly in the mirror-image case.)

When can you kill the electric field? By the identical argument with the roles swapped:

$$B^2 - \frac{E^2}{c^2} > 0 \quad (\text{i.e. } E < cB), \quad \text{and} \quad \mathbf{E} \cdot \mathbf{B} = 0. \quad (2.6.62)$$

Worked Example 2 does this case in full and finds the frame velocity, which turns out to be the drift velocity every mass spectrometer relies on.

And if $\mathbf{E} \cdot \mathbf{B} \neq 0$, neither field can be removed in any frame, ever. The two fields are then irreducibly both present, and the best you can do is find a frame where they are parallel.

READING THE INVARIANTS AS A CLASSIFICATION

The pair $(F_{\mu\nu}F^{\mu\nu}, \epsilon FF)$ classifies field configurations into frame-independent types, exactly as the sign of Δs^2 classified pairs of events in Chapter 2.3 into timelike, spacelike and null.

INVARIANTS	NAME	SIMPLEST FRAME
$E > cB, \mathbf{E} \cdot \mathbf{B} = 0$	electric	a frame with $\mathbf{B} = \mathbf{0}$: pure electrostatics
$E < cB, \mathbf{E} \cdot \mathbf{B} = 0$	magnetic	a frame with $\mathbf{E} = \mathbf{0}$: pure magnetostatics
$E = cB, \mathbf{E} \cdot \mathbf{B} = 0$	null	none simpler — a radiation field
$\mathbf{E} \cdot \mathbf{B} \neq 0$	generic	a frame with $\mathbf{E} \parallel \mathbf{B}$

The null case is the boundary between the other two and is exactly the case that cannot be simplified — the same way a null interval sits on the boundary of the light cone and cannot be transformed into either a pure time separation or a pure space separation. Light is the null case of electromagnetism in precisely the sense that a light ray is the null case of a worldline.

8 · The Lorentz force, covariantly

Chapter 2.5 §6.1 established the shape any relativistic force law must have:

$$\frac{dp^\mu}{d\tau} = f^\mu, \quad \text{with the constraint} \quad f \cdot u = 0. \quad (2.6.63)$$

What can f^μ be for a charge in an electromagnetic field? It must be a four-vector, linear in the charge, and built from the only field object available. With u^μ the particle's four-velocity there is essentially one candidate:

$$\boxed{\frac{dp^\mu}{d\tau} = q F^{\mu\nu} u_\nu.} \quad (2.6.64)$$

Chapter 2.4's grind box on the quotient theorem already flagged this: requiring $F^{\mu\nu} u_\nu$ to be a four-vector for *every* four-velocity is what certifies $F^{\mu\nu}$ as a tensor. Now we verify that (2.6.64) is the force law you already know.

8.1 · The spatial components

Lower the index on the four-velocity $u^\mu = \gamma(c, \mathbf{v})$ of Chapter 2.5 §1.3, which flips the sign of the spatial part:

$$u_\nu = \eta_{\nu\alpha} u^\alpha = \gamma(c, -\mathbf{v}). \quad (2.6.65)$$

Take $\mu = i$ and split the sum over ν into its time and space parts:

$$\begin{aligned}
 qF^{i\nu}u_\nu &= q[F^{i0}u_0 + F^{ij}u_j] \\
 &= q\left[\frac{E^i}{c}(\gamma c) + (-\epsilon^{ijk}B^k)(-\gamma v^j)\right] \\
 &= \gamma q[E^i + \epsilon^{ijk}v^j B^k] = \gamma q[\mathbf{E} + \mathbf{v} \times \mathbf{B}]^i.
 \end{aligned} \tag{2.6.66}$$

The left-hand side is $dp^i/d\tau = \gamma dp^i/dt$, since $dt/d\tau = \gamma$ (Chapter 2.5 §1.2). Cancel the common γ :

$$\boxed{\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})}. \tag{2.6.67}$$

The Lorentz force law — the one Chapter 1.1 §3.3 had to quote, now derived as the spatial part of a four-vector equation. Notice that $\mathbf{p} = \gamma m\mathbf{v}$ here, not $m\mathbf{v}$; the relativistic correction rides along for free.

8.2 · The time component is the power

Take $\mu = 0$. The $\nu = 0$ term dies because $F^{00} = 0$:

$$qF^{0\nu}u_\nu = qF^{0j}u_j = q\left(-\frac{E^j}{c}\right)(-\gamma v^j) = \frac{\gamma q}{c}\mathbf{E} \cdot \mathbf{v}. \tag{2.6.68}$$

And the left-hand side is $dp^0/d\tau = \gamma d(E/c)/dt$, where E is now the particle's energy of Chapter 2.5 §3 (an unfortunate clash of symbols that the whole subject lives with). Cancelling γ/c :

$$\frac{dE}{dt} = q\mathbf{E} \cdot \mathbf{v}. \tag{2.6.69}$$

The magnetic field does no work. That familiar fact is here not a separate observation but a consequence of where $\mathbf{B}$ sits in the matrix: the F^{0j} entries are purely electric, so only $\mathbf{E}$ can appear in the energy equation. And (2.6.69) is Chapter 2.5's work-energy theorem $dE/dt = \mathbf{F} \cdot \mathbf{v}$ with $\mathbf{F}$ the Lorentz force, since $(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} = 0$ automatically.

8.3 · The constraint is automatic, and why

Chapter 2.5 §6.1 proved that any four-force must satisfy $f \cdot u = 0$, because $u \cdot u = c^2$ is constant. That is a strong condition, and it is not obvious that an arbitrary guess for f^μ would satisfy it. Check (2.6.64):

$$f \cdot u = f^\mu u_\mu = qF^{\mu\nu}u_\nu u_\mu = 0. \tag{2.6.70}$$

Why zero? Because $F^{\mu\nu}$ is antisymmetric under $\mu \leftrightarrow \nu$ while $u_\mu u_\nu$ is symmetric, and *a symmetric object fully contracted with an antisymmetric one vanishes*. The argument, once, in case it has not been made explicit before: relabel the dummy indices $\mu \leftrightarrow \nu$ in the sum, which changes nothing;

$$S \equiv F^{\mu\nu} u_\mu u_\nu = F^{\nu\mu} u_\nu u_\mu = -F^{\mu\nu} u_\mu u_\nu = -S \quad \implies \quad S = 0. \quad (2.6.71)$$

THE STRUCTURAL CHECK THAT HAD TO WORK

Look at what just happened. The constraint $f \cdot u = 0$ is a statement about the geometry of spacetime — it says a four-force can only push a four-velocity *sideways*, because four-velocities all have the same length. It came from Chapter 2.5 with no reference to electromagnetism whatsoever.

And it is satisfied by (2.6.64) **because $F^{\mu\nu}$ is antisymmetric** — a fact that came from $F = \partial A - \partial A$, with no reference to spacetime geometry whatsoever. Two independent constraints, from two unrelated directions, matching exactly. Its physical content is (2.6.69): the reason the magnetic force does no work is the reason a four-force is orthogonal to a four-velocity. Had $F^{\mu\nu}$ had any symmetric part, charges would gain energy from nothing.

9 · The action

Chapter 1.2 §8.1 listed eight actions and promised each would be constructed in its own chapter. Line three said *electromagnetic field, Chapter 2.6*. Here it is, with the source term and the SI constants restored:

$$\boxed{\mathcal{L}_{\text{EM}} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - j_\mu A^\mu.} \quad (2.6.72)$$

Two terms, one line, and it contains every result of §3. The claim to verify is that feeding (2.6.72) to the field Euler–Lagrange equation of Chapter 1.2 §8 returns $\partial_\mu F^{\mu\nu} = \mu_0 j^\nu$.

9.1 · Varying it

The dynamical variable is A_ν — four fields, one for each value of ν — so the Euler–Lagrange equation carries a free index:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0. \quad (2.6.73)$$

The second term is immediate: only $-j_\mu A^\mu = -j^\nu A_\nu$ contains A_ν undifferentiated, so $\partial \mathcal{L} / \partial A_\nu = -j^\nu$. The first term is the work, and the grind box does it slowly. The answer is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{\mu_0} F^{\mu\nu}. \quad (2.6.74)$$

Substituting both into (2.6.73):

$$\partial_\mu \left(-\frac{1}{\mu_0} F^{\mu\nu} \right) + j^\nu = 0 \quad \implies \quad \partial_\mu F^{\mu\nu} = \mu_0 j^\nu. \quad (2.6.75)$$

Which is (2.6.19) — Gauss's law and Ampère–Maxwell, both of them, out of one scalar. The homogeneous pair needs no varying at all: it holds identically the moment you write F in terms of A , which is §3.4.

▼ Grind box — differentiating $F_{\alpha\beta} F^{\alpha\beta}$ with respect to $\partial_\mu A_\nu$

This is pure index gymnastics and it is worth doing once with every step visible, because the same manipulation reappears in Chapters 5.2, 6.3 and 6.4 with more indices.

Step 1 — the chain rule on a product. Write $\mathcal{L}_F = -\frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta}$ and note $F_{\alpha\beta} F^{\alpha\beta} = \eta^{\alpha\gamma} \eta^{\beta\delta} F_{\alpha\beta} F_{\gamma\delta}$, so it is quadratic in the $F_{\alpha\beta}$ with constant coefficients. Hence

$$\frac{\partial}{\partial(\partial_\mu A_\nu)} (F_{\alpha\beta} F^{\alpha\beta}) = 2F^{\alpha\beta} \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)}.$$

Step 2 — the inner derivative. Since $F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha$ and the sixteen quantities $\partial_\mu A_\nu$ are treated as independent variables,

$$\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} = \delta^\mu_\alpha \delta^\nu_\beta \quad \implies \quad \frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} = \delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha.$$

The two deltas in each product are what turn "differentiate with respect to a particular component" into "select that component"; the two terms are the two places ∂A appears in F .

Step 3 — contract. The deltas rename indices:

$$2F^{\alpha\beta} (\delta^\mu_\alpha \delta^\nu_\beta - \delta^\mu_\beta \delta^\nu_\alpha) = 2(F^{\mu\nu} - F^{\nu\mu}) = 4F^{\mu\nu},$$

the last step by antisymmetry — and there is the factor of 4 that the $\frac{1}{4}$ in (2.6.72) was put there to cancel. So

$$\frac{\partial \mathcal{L}_F}{\partial(\partial_\mu A_\nu)} = -\frac{1}{4\mu_0} \cdot 4F^{\mu\nu} = -\frac{1}{\mu_0} F^{\mu\nu},$$

which is (2.6.74). The source term contains no derivatives of A and contributes nothing here. ✓

Why the sign of the whole Lagrangian is what it is. Expand (2.6.72) into three-vector language using (2.6.56):

$$-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} = -\frac{1}{2\mu_0} \left(B^2 - \frac{E^2}{c^2} \right) = \frac{\epsilon_0 E^2}{2} - \frac{B^2}{2\mu_0}.$$

That is *kinetic minus potential* — the electric term carries the time derivatives of $\mathbf{A}$ and plays the role of T , the magnetic term carries the spatial derivatives and plays the role of V . The overall sign of

(2.6.72) is fixed by demanding that T come out positive, which is Chapter 1.2 §5's discussion transplanted verbatim to a field. Get it wrong and the field has negative kinetic energy, which Chapter 5.2 shows is a catastrophe rather than a sign convention.

9.2 · Why this Lagrangian and not another

The variation above is a verification, not a derivation: we wrote down (2.6.72) and checked it. So where did it come from? The honest answer is a short list of demands.

1. **Lorentz invariance.** $\mathcal{L}$ must be a scalar, or the action is not the same number for every observer and the theory has a preferred frame. That immediately restricts us to contractions: $F_{\mu\nu}F^{\mu\nu}$, $\epsilon F F$, $A_\mu A^\mu$, $j_\mu A^\mu$, and products and derivatives of these.
2. **Gauge invariance.** $\mathcal{L}$ must be unchanged (up to a four-divergence) under (2.6.29). This kills $A_\mu A^\mu$ outright — it is not gauge invariant, and it would have been a photon mass term. *The photon is massless because a mass term is not gauge invariant.* It permits $j_\mu A^\mu$ only because charge is conserved, as §4.1 showed.
3. **Second-order equations of motion.** $\mathcal{L}$ should contain at most first derivatives of A , so that Euler–Lagrange returns something second order and the initial-value problem is the usual one.
4. **Parity, if you want it.** $\epsilon_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$ passes tests 1–3 but is a pseudoscalar, and moreover is a total derivative, so it does not affect the classical field equations at all.

What survives at lowest order is $F_{\mu\nu}F^{\mu\nu}$ and $j_\mu A^\mu$, with two constants — one of which is fixed by matching Coulomb's law and the other absorbed into the normalisation of A . **The form is very nearly forced.**

THREE DOORS THIS OPENS

Chapter 6.4 — Yang–Mills. Take (2.6.72), let A^μ carry an internal index and become matrix-valued, and define $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$. Then write down $-\frac{1}{2} \text{tr } F_{\mu\nu}F^{\mu\nu}$ — *the same expression*, entry seven in Chapter 1.2's table. The commutator is the only new ingredient, and it is what makes gluons carry colour charge and interact with each other. The strong and weak interactions are this chapter with a non-commuting symmetry group.

Chapter 5.9 — effective field theory. The argument of §9.2 is the beginning of a general method: *list the fields, list the symmetries, write down every term allowed, and order them by how many derivatives they have.* The leading term is the theory; the rest are corrections suppressed by powers of energy. Electromagnetism looks fundamental partly because the leading allowed term is so simple. Higher terms exist — $(F_{\mu\nu}F^{\mu\nu})^2$ is the Euler–Heisenberg term, which makes light scatter off light — and are suppressed by four powers of the electron mass, which is why nobody noticed for a century.

Chapter 5.2 — quantisation. A Lagrangian is the input to a path integral. Once you have (2.6.72), quantising the electromagnetic field is a well-posed problem, and its answer is the photon. The vertex of quantum electrodynamics is the term $-j_\mu A^\mu$ (Chapter 5.6), which is the entire interaction between light and matter and which you have just written down.

A NOTE ON UNITS, BEFORE PART V

The factors of μ_0 , ϵ_0 and c cluttering this chapter are SI, and they carry real information here: they are what let you compare a derived force against a laboratory measurement in newtons, which §6 and the worked examples do. From Part V onward the book switches to **natural units**, $\hbar = c = 1$, in the Heaviside–Lorentz convention where the 4π is moved out of Coulomb's law and $\epsilon_0 = \mu_0 = 1$. In those units (2.6.72) reads $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - j_\mu A^\mu$, which is how you will see it written everywhere else, and every equation in this chapter loses its decoration. Nothing physical changes; the constants can always be restored by dimensional analysis. We keep them here because Part II is about measurable consequences and dropping c while still learning what c means would be perverse.

10 · Field momentum, and Newton's third law repaired

Chapter 1.1 §3.3 put two moving charges alone in the universe and showed that the total force on them does not vanish. Momentum appeared from nowhere at a computable rate. That chapter offered two escapes — momentum conservation is false, or momentum is stored somewhere that is not a particle — took the second, named the electromagnetic field as the only candidate, and left an IOU. This section pays it, and the payment is exact.

10.1 · The stress-energy tensor, from Noether

Chapter 1.4 proved that every continuous symmetry yields a conserved current. The symmetry we have not yet used is the most basic one there is: **the laws are the same here as there, and now as then** — invariance under spacetime translations $x^\mu \rightarrow x^\mu + a^\mu$. Since a^μ has four components, we get four conserved currents, one per direction, and they assemble into a rank-2 object.

Run Chapter 1.4 §6's machinery. Under a translation the field changes by $\delta A_\lambda = -a^\nu \partial_\nu A_\lambda$ and the Lagrangian density by $\delta \mathcal{L} = -a^\nu \partial_\nu \mathcal{L} = \partial_\mu (-a^\nu \delta^\mu_\nu \mathcal{L})$ — a four-divergence, which is exactly the loosened invariance condition of Chapter 1.4 §1.2. The Noether current for the ν -th translation is therefore

$$T_{\text{can}}^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\lambda)} \partial^\nu A_\lambda - \eta^{\mu\nu} \mathcal{L} = -\frac{1}{\mu_0} F^{\mu\lambda} \partial^\nu A_\lambda + \frac{\eta^{\mu\nu}}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta}, \quad (2.6.76)$$

using (2.6.74) and setting $j^\mu = 0$ (we are asking what the free field carries). It satisfies $\partial_\mu T_{\text{can}}^{\mu\nu} = 0$ by Noether's theorem.

And it is wrong in two ways: it is not symmetric in $\mu\nu$, and it is not gauge invariant — it contains A_λ bare, so different gauges assign different energy densities to the same physical field. Both defects are fixed by the freedom Chapter 1.4's grind box identified: adding $\partial_\lambda \Sigma^{\lambda\mu\nu}$ with Σ antisymmetric in its first two indices changes no conserved charge. Take

$$\Sigma^{\lambda\mu\nu} = \frac{1}{\mu_0} F^{\mu\lambda} A^\nu, \quad \text{antisymmetric in } \lambda\mu \text{ because } F \text{ is.} \quad (2.6.77)$$

Its divergence is, using $\partial_\lambda F^{\mu\lambda} = -\partial_\lambda F^{\lambda\mu} = -\mu_0 j^\mu = 0$ in the source-free region,

$$\partial_\lambda \Sigma^{\lambda\mu\nu} = \frac{1}{\mu_0} \left[\underbrace{(\partial_\lambda F^{\mu\lambda})}_{=0} A^\nu + F^{\mu\lambda} \partial_\lambda A^\nu \right] = \frac{1}{\mu_0} F^{\mu\lambda} \partial_\lambda A^\nu. \quad (2.6.78)$$

Add it to (2.6.76) and the two ∂A terms combine into an F :

$$T^{\mu\nu} = -\frac{1}{\mu_0} F^{\mu\lambda} \underbrace{(\partial^\nu A_\lambda - \partial_\lambda A^\nu)}_{= F^\nu{}_\lambda} + \frac{\eta^{\mu\nu}}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta}. \quad (2.6.79)$$

Since $F^\nu{}_\lambda = -F_\lambda{}^\nu$, this is

$$\boxed{T^{\mu\nu} = \frac{1}{\mu_0} \left(F^{\mu\lambda} F_\lambda{}^\nu + \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right)}. \quad (2.6.80)$$

Now it is manifestly gauge invariant (only F appears), and it is symmetric — check: the first term is $\eta_{\lambda\rho} F^{\mu\lambda} F^{\rho\nu}$, and swapping $\mu \leftrightarrow \nu$ together with the dummy relabel $\lambda \leftrightarrow \rho$ returns it, the two minus signs from antisymmetry cancelling. Symmetry is not cosmetic: Chapter 3.6 puts $T^{\mu\nu}$ on the right-hand side of Einstein's equation, whose left-hand side is symmetric by construction.

10.2 · What its components are

T^{00} . Only λ spatial contributes to $F^{0\lambda} F_\lambda{}^0$, and lowering the index costs a minus:

$$F^{0\lambda} F_\lambda{}^0 = F^{0i} \eta_{ij} F^{j0} = -F^{0i} F^{i0} = + \sum_i (F^{0i})^2 = \frac{E^2}{c^2}, \quad (2.6.81)$$

using $F^{i0} = -F^{0i}$ and $F^{0i} = -E^i/c$. With $\eta^{00} = 1$ and (2.6.56) for the second term,

$$T^{00} = \frac{1}{\mu_0} \left[\frac{E^2}{c^2} + \frac{1}{4} \cdot 2 \left(B^2 - \frac{E^2}{c^2} \right) \right] = \frac{E^2}{2\mu_0 c^2} + \frac{B^2}{2\mu_0} = \boxed{\frac{\epsilon_0 E^2}{2} + \frac{B^2}{2\mu_0}} \quad (2.6.82)$$

using $1/(\mu_0 c^2) = \epsilon_0$. **That is the energy density of the electromagnetic field** — the expression you have probably met as a stated fact, here derived as the conserved Noether charge density of time translation, which is what "energy" has meant since Chapter 1.4 §3.1.

T^{0i} . Now $\eta^{0i} = 0$, so only the first term survives:

$$\begin{aligned} F^{0\lambda} F_\lambda{}^i &= F^{0j} \eta_{jk} F^{ki} = -F^{0j} F^{ji} = +F^{0j} F^{ij} \\ &= \left(-\frac{E^j}{c} \right) (-\epsilon^{ijk} B^k) = \frac{1}{c} \epsilon^{ijk} E^j B^k. \end{aligned} \quad (2.6.83)$$

Only $\lambda = j$ contributes, since $F^{00} = 0$; the metric η_{jk} supplies one minus sign and the antisymmetry $F^{ji} = -F^{ij}$ supplies the other. Hence

$$T^{0i} = \frac{1}{\mu_0 c} \epsilon^{ijk} E^j B^k = \frac{1}{c} \left[\frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \right]^i = \frac{S^i}{c}, \quad \mathbf{S} \equiv \frac{\mathbf{E} \times \mathbf{B}}{\mu_0}. \quad (2.6.84)$$

$\mathbf{S}$ is the **Poynting vector**. And since $T^{0\mu}/c$ is by construction the density of p^μ — the 0-index labels the conserved density, the μ labels which component of momentum — the momentum density of the field is

$$\mathbf{g} = \frac{T^{0i}}{c} \hat{\mathbf{e}}_i = \frac{\mathbf{S}}{c^2} = \epsilon_0 \mathbf{E} \times \mathbf{B}. \quad (2.6.85)$$

Which is precisely the expression Chapter 1.1 promised, arriving here as a component of a Noether current rather than as an assertion.

10.3 · The conservation law, with sources

Now restore the charges, because it is the exchange between field and matter that we care about. Take the divergence of (2.6.80):

$$\mu_0 \partial_\mu T^{\mu\nu} = \underbrace{(\partial_\mu F^{\mu\lambda}) F_\lambda^\nu}_{(1)} + \underbrace{F^{\mu\lambda} \partial_\mu F_\lambda^\nu}_{(2)} + \underbrace{\frac{1}{2} F^{\alpha\beta} \partial^\nu F_{\alpha\beta}}_{(3)}. \quad (2.6.86)$$

Term (1) is the field equation: $\partial_\mu F^{\mu\lambda} = \mu_0 j^\lambda$, so it is $\mu_0 j^\lambda F_\lambda^\nu$. Terms (2) and (3) cancel identically, by the Bianchi identity — the grind box does it. So

$$\partial_\mu T^{\mu\nu} = j_\lambda F^{\lambda\nu} = -F^{\nu\lambda} j_\lambda. \quad (2.6.87)$$

The right-hand side is not zero, and it should not be: the field is exchanging energy and momentum with the charges. Expand it. For $\nu = i$, using $j_0 = c\rho$ and $j_j = -J^j$,

$$F^{i\lambda} j_\lambda = \frac{E^i}{c} (c\rho) + (-\epsilon^{ijk} B^k) (-J^j) = [\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}]^i, \quad (2.6.88)$$

which is the **Lorentz force per unit volume** — (2.6.67) written for a continuous distribution. Meanwhile the left-hand side, with $\partial_0 = \frac{1}{c} \partial_t$ and $T^{0i} = c g^i$, is $\partial_t g^i + \partial_j T^{ji}$. Therefore

$$\frac{\partial \mathbf{g}}{\partial t} + \nabla \cdot \mathbf{T} = -(\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) = -\mathbf{f}_{\text{mech}}, \quad (2.6.89)$$

with $\mathbf{T}$ the spatial block T^{ij} (the *Maxwell stress tensor*). Integrate over all space. The middle term becomes a surface integral at infinity by the divergence theorem, and for a system of charges confined to a finite region the fields fall off

at least as fast as $1/r^2$, so $T^{ij} \sim 1/r^4$ while the area grows as r^2 : the surface term vanishes. The last term integrates to the total force on the matter, which is $d\mathbf{P}_{\text{mech}}/dt$. Hence

$$\boxed{\frac{d}{dt}(\mathbf{P}_{\text{mech}} + \mathbf{P}_{\text{field}}) = \mathbf{0}, \quad \mathbf{P}_{\text{field}} = \epsilon_0 \int \mathbf{E} \times \mathbf{B} \, d^3x.} \quad (2.6.90)$$

The $\nu = 0$ component of (2.6.87) gives, by the identical route, $\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$ — **Poynting's theorem**: the field energy in a region falls by the flux of $\mathbf{S}$ out of it plus the work done on the charges inside. The two statements are one four-vector equation.

▼ Grind box — why terms (2) and (3) cancel

Write term (2) with cleaner dummy names, $\mu \rightarrow \alpha$ and $\lambda \rightarrow \beta$:

$$(2) = F^{\alpha\beta} \partial_\alpha F_\beta^\nu.$$

Because $F^{\alpha\beta}$ is antisymmetric, only the antisymmetric part of whatever it multiplies survives. So we may replace the second factor by half of it minus its $\alpha \leftrightarrow \beta$ swap:

$$(2) = \frac{1}{2} F^{\alpha\beta} (\partial_\alpha F_\beta^\nu - \partial_\beta F_\alpha^\nu).$$

Now the Bianchi identity (2.6.25), with the third index raised (legitimate: raise it on every term with the same η):

$$\partial_\alpha F_\beta^\nu + \partial_\beta F_\alpha^\nu + \partial^\nu F_{\alpha\beta} = 0.$$

Use $F_{\alpha}^\nu = -F_{\alpha}^\nu$ on the middle term and rearrange:

$$\partial_\alpha F_\beta^\nu - \partial_\beta F_\alpha^\nu = -\partial^\nu F_{\alpha\beta}.$$

Substitute:

$$(2) = -\frac{1}{2} F^{\alpha\beta} \partial^\nu F_{\alpha\beta} = -(3). \quad \blacksquare$$

Note what did the work: the *homogeneous* half of Maxwell — the half §3.4 called bookkeeping. It is not decorative after all. Without it $T^{\mu\nu}$ would not be conserved, and energy and momentum would not balance.

10.4 · Back to Chapter 1.1

The configuration, unchanged: charge q_1 at the origin moving with $\mathbf{v}_1 = v\hat{\mathbf{x}}$; charge q_2 at $\mathbf{r}_2 = d\hat{\mathbf{x}}$ — directly ahead of it — moving with $\mathbf{v}_2 = v\hat{\mathbf{y}}$. Both in uniform motion at the instant considered.

What 1.1 found. Charge 1 produces no magnetic field directly ahead of itself, so charge 2 feels no magnetic force; charge 2 does produce one at charge 1's location, so charge 1 does. The magnetic residue was

$$(\mathbf{F}_{12} + \mathbf{F}_{21})_{\text{mag}} = -\frac{\mu_0 q_1 q_2 v^2}{4\pi d^2} \hat{\mathbf{y}}. \quad (2.6.91)$$

What 1.1 was entitled to drop, and we are not. That chapter said the electric forces "are equal and opposite along the line joining them, so they cancel". True to zeroth order in v/c — and false at order v^2/c^2 , which is the very order the magnetic residue lives at. Chapter 1.1 was making a qualitative point and the omission cost it nothing; we are about to make a quantitative one, so we need the correction. It is three lines with §5's transformation rules.

Charge 1's field at charge 2. Boost to charge 1's rest frame S' . The field point ($t = 0$, $x = d$) has $x' = \gamma(d - v \cdot 0) = \gamma d$, so in S' the charge sits at a Coulomb distance γd and the field there is $q_1 / (4\pi\epsilon_0 \gamma^2 d^2)$, pointing along $\hat{\mathbf{x}}$ — parallel to the boost. By §5, $E_{\parallel}$ is unchanged. So

$$\mathbf{E}_1(\mathbf{r}_2) = \frac{q_1}{4\pi\epsilon_0 d^2} \frac{1}{\gamma^2} \hat{\mathbf{x}}. \quad (2.6.92)$$

Charge 2's field at charge 1. Boost to charge 2's rest frame, which moves along $\hat{\mathbf{y}}$. The displacement from charge 2 to the origin, $-d\hat{\mathbf{x}}$, is *perpendicular* to that boost, so it is unchanged and the Coulomb distance is d . The field is $-q_2 / (4\pi\epsilon_0 d^2) \hat{\mathbf{x}}$ there, and $\hat{\mathbf{x}}$ is perpendicular to the boost, so by §5 it is multiplied by γ on transforming to the lab:

$$\mathbf{E}_2(\mathbf{r}_1) = -\frac{q_2 \gamma}{4\pi\epsilon_0 d^2} \hat{\mathbf{x}}. \quad (2.6.93)$$

The two electric forces are $q_2 \mathbf{E}_1(\mathbf{r}_2)$ and $q_1 \mathbf{E}_2(\mathbf{r}_1)$, and they no longer cancel: one is weakened by γ^{-2} and the other *strengthened* by γ , because charge 1 is moving along the line joining them and charge 2 is moving across it. Their sum, expanded to order β^2 with $\gamma^{-2} = 1 - \beta^2$ and $\gamma \approx 1 + \frac{1}{2}\beta^2$:

$$\begin{aligned} (\mathbf{F}_{12} + \mathbf{F}_{21})_{\text{elec}} &= \frac{q_1 q_2}{4\pi\epsilon_0 d^2} \left(\frac{1}{\gamma^2} - \gamma \right) \hat{\mathbf{x}} \\ &\approx \frac{q_1 q_2}{4\pi\epsilon_0 d^2} \left(-\frac{3}{2}\beta^2 \right) \hat{\mathbf{x}} = -\frac{3}{2} \frac{\mu_0 q_1 q_2 v^2}{4\pi d^2} \hat{\mathbf{x}}, \end{aligned} \quad (2.6.94)$$

using $\beta^2/\epsilon_0 = v^2/(c^2\epsilon_0) = \mu_0 v^2$. So the third law fails in the $\hat{\mathbf{x}}$ direction too, and by a *larger* amount than in $\hat{\mathbf{y}}$. The total rate at which mechanical momentum is being created is

$$\frac{d\mathbf{P}_{\text{mech}}}{dt} = -\frac{\mu_0 q_1 q_2 v^2}{4\pi d^2} \left(\frac{3}{2} \hat{\mathbf{x}} + \hat{\mathbf{y}} \right). \quad (2.6.95)$$

10.5 · And where it went

Now compute the field momentum. To the order we need, each charge's magnetic field is $\mathbf{B}_a = \mathbf{v}_a \times \mathbf{E}_a / c^2$ (which §5 gives exactly, by boosting a rest frame where $\mathbf{B} = \mathbf{0}$), and each $\mathbf{E}_a$ may be taken as Coulomb, since $\mathbf{E} \times \mathbf{B}$ is already

first order in v/c and corrections to $\mathbf{E}$ are second. Expand (2.6.90):

$$\mathbf{P}_{\text{field}} = \epsilon_0 \int (\mathbf{E}_1 + \mathbf{E}_2) \times (\mathbf{B}_1 + \mathbf{B}_2) d^3x. \quad (2.6.96)$$

The two *self* terms $\mathbf{E}_a \times \mathbf{B}_a$ depend only on $\mathbf{v}_a$, which is constant, so however large they are (and they are infinite — see the caution below) they contribute nothing to $d\mathbf{P}/dt$. Keep the cross terms, and use $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$ on each:

$$\mathbf{P}_{\text{field}}^{\text{int}} = \frac{\epsilon_0}{c^2} \int [(\mathbf{v}_1 + \mathbf{v}_2)(\mathbf{E}_1 \cdot \mathbf{E}_2) - \mathbf{E}_2(\mathbf{E}_1 \cdot \mathbf{v}_2) - \mathbf{E}_1(\mathbf{E}_2 \cdot \mathbf{v}_1)] d^3x. \quad (2.6.97)$$

Two integrals are needed, and both are elementary (grind box). With $\mathbf{d} = \mathbf{r}_2 - \mathbf{r}_1$, $d = |\mathbf{d}|$ and $\hat{\mathbf{n}} = \mathbf{d}/d$:

$$\int \mathbf{E}_1 \cdot \mathbf{E}_2 d^3x = \frac{q_1 q_2}{4\pi\epsilon_0^2 d}, \quad \int E_1^i E_2^j d^3x = \frac{q_1 q_2}{8\pi\epsilon_0^2 d} (\delta^{ij} - n^i n^j). \quad (2.6.98)$$

Substitute. Write $\mathbf{V} \equiv \mathbf{v}_1 + \mathbf{v}_2$; the two tensor terms combine, using the symmetry of $M^{ij} \equiv \int E_1^i E_2^j d^3x$, into $M^{ij} V^j$, and the prefactors collapse with $1/(\epsilon_0 c^2) = \mu_0$:

$$\begin{aligned} \mathbf{P}_{\text{field}}^{\text{int}} &= \frac{\epsilon_0}{c^2} \cdot \frac{q_1 q_2}{4\pi\epsilon_0^2 d} \left[\mathbf{V} - \frac{1}{2} (\mathbf{V} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{V})) \right] \\ &= \boxed{\frac{\mu_0 q_1 q_2}{8\pi d} [\mathbf{V} + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{V})]}. \end{aligned} \quad (2.6.99)$$

For the configuration at hand, at the instant $t = 0$: $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, $\mathbf{V} = v(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, $\hat{\mathbf{n}} \cdot \mathbf{V} = v$, so

$$\mathbf{P}_{\text{field}} = \frac{\mu_0 q_1 q_2 v}{8\pi d} (2\hat{\mathbf{x}} + \hat{\mathbf{y}}) \neq \mathbf{0}. \quad (2.6.100)$$

There is momentum in the empty space between the charges. Now differentiate. The separation vector evolves as $\mathbf{d}(t) = (d - vt)\hat{\mathbf{x}} + vt\hat{\mathbf{y}}$, since charge 1 moves $+x$ and charge 2 moves $+y$; at $t = 0$ this gives $\dot{\mathbf{d}} = -v\hat{\mathbf{x}} + v\hat{\mathbf{y}}$. Grinding through the product rule (grind box):

$$\left. \frac{d\mathbf{P}_{\text{field}}}{dt} \right|_{t=0} = \frac{\mu_0 q_1 q_2 v^2}{4\pi d^2} \left(\frac{3}{2}\hat{\mathbf{x}} + \hat{\mathbf{y}} \right). \quad (2.6.101)$$

Compare with (2.6.95). Term by term, including the awkward $\frac{3}{2}$ that Chapter 1.1 never saw:

$$\boxed{\frac{d\mathbf{P}_{\text{mech}}}{dt} + \frac{d\mathbf{P}_{\text{field}}}{dt} = \mathbf{0}.} \quad (2.6.102)$$

THE LONGEST-RUNNING PROMISE IN THE BOOK, CLOSED

Chapter 1.1 §3.3 wrote down a momentum non-conservation of $-\mu_0 q_1 q_2 v^2 \hat{\mathbf{y}} / 4\pi d^2$ and offered you a choice: abandon momentum conservation, or accept that the field is a physical object with momentum of its own. The second was right, and now it is *computed*: the missing momentum is (2.6.100), it lives in the field, and its rate of change is exactly minus the rate at which the particles are gaining momentum from nowhere — in both components, with the $\frac{3}{2}$ that only shows up if you are honest about the electric forces at order v^2/c^2 .

Notice how the repair works. Newton's third law presumed that all the momentum in the universe is carried by particles and that they exchange it instantaneously. Relativity forbids the second clause, and the first then cannot survive: if charge 1 moves now and charge 2 will not know for d/c seconds, the momentum has to be *somewhere* during the interval. The third law is not a fundamental principle that electromagnetism violates. It is a low-velocity approximation to a four-vector conservation law, and it fails exactly where the approximation does.

The conceptual bill is large and worth paying explicitly. A field can no longer be read as a bookkeeping device — a table of where the force would be if you put a test charge there. Something that stores energy, stores momentum, exerts stresses, and can carry momentum across a room during the interval when neither particle has it, is not a table. It is a physical system with its own degrees of freedom and its own dynamics. That is why Part V quantises it, and why the resulting quanta — photons — are as real as electrons rather than being a way of talking about electrons.

△ TWO HONEST CAUTIONS

The self terms are infinite. $\epsilon_0 \int \mathbf{E}_a \times \mathbf{B}_a d^3x$ diverges at the location of a point charge, as does the self-energy $\frac{\epsilon_0}{2} \int E_a^2$. We dodged it above by noting that these terms are constant when $\mathbf{v}_a$ is, so they drop out of the rate. That dodge is legitimate here and evasive in general: the divergence is the classical electron self-energy problem, it is genuinely unsolved in classical physics, and it is the ancestor of the renormalisation programme of Chapter 5.9. Point charges are an idealisation and they bite.

What we computed and what we did not. The general theorem (2.6.90) is exact, derived from $\partial_\mu T^{\mu\nu}$ with no approximation. The two-charge *check* keeps the velocity-dependent terms to order v^2/c^2 and drops the acceleration (radiation) fields, whose contribution is balanced separately against the $\dot{\mathbf{v}}$ terms in (2.6.99) — a second, independent piece of bookkeeping controlled by a different small parameter, the ratio of the charges' Coulomb energy to their kinetic energy. Nothing was assumed to make (2.6.102) come out; every coefficient was computed and then compared.

▼ Grind box — the two field integrals of (2.6.98)

Both are done with the same two moves: integrate by parts, then use Gauss's law $\nabla \cdot \mathbf{E}_a = \rho_a/\epsilon_0$ with $\rho_a = q_a \delta^3(\mathbf{x} - \mathbf{r}_a)$. Surface terms vanish because the fields fall off as $1/r^2$ and the integrands as $1/r^3$ or faster.

The scalar one. Write $\mathbf{E}_1 = -\nabla\phi_1$ and integrate by parts:

$$\int \mathbf{E}_1 \cdot \mathbf{E}_2 \, d^3x = -\int \nabla\phi_1 \cdot \mathbf{E}_2 \, d^3x = \int \phi_1 (\nabla \cdot \mathbf{E}_2) \, d^3x = \frac{q_2}{\epsilon_0} \phi_1(\mathbf{r}_2) = \frac{q_1 q_2}{4\pi\epsilon_0^2 d}.$$

(Multiplied by ϵ_0 this is the familiar interaction energy $q_1 q_2 / 4\pi\epsilon_0 d$ — a useful check that the normalisation is right.)

The tensor one. Call it $M^{ij}(\mathbf{d})$. Three facts pin it down.

(i) *It is symmetric.* Invert through the midpoint, $\mathbf{x} \rightarrow \mathbf{r}_1 + \mathbf{r}_2 - \mathbf{x}$. Under that map $\mathbf{x} - \mathbf{r}_1 \mapsto -(\mathbf{x} - \mathbf{r}_2)$, so $\mathbf{E}_1 \mapsto -(q_1/q_2)\mathbf{E}_2$ and $\mathbf{E}_2 \mapsto -(q_2/q_1)\mathbf{E}_1$, and the Jacobian is 1. Hence $M^{ij} = M^{ji}$.

(ii) *Its form.* The only vectors in the problem are $\hat{\mathbf{n}}$ and the coordinate axes, so by rotational symmetry about $\hat{\mathbf{n}}$ and (i),

$$M^{ij} = \frac{A}{d} \delta^{ij} + \frac{B}{d} n^i n^j,$$

with A, B constants — the $1/d$ because M has dimensions of the scalar integral, which scales as $1/d$.

(iii) *Two equations.* The trace is the scalar integral just computed:

$$3A + B = \frac{q_1 q_2}{4\pi\epsilon_0^2}.$$

For the second, differentiate with respect to $\mathbf{d}$ and contract. Since $\mathbf{E}_2$ depends on $\mathbf{x} - \mathbf{r}_2$ and $\mathbf{r}_2 = \mathbf{r}_1 + \mathbf{d}$, we have $\partial E_2^j / \partial d^k = -\partial_k E_2^j$, so

$$\frac{\partial M^{ij}}{\partial d^j} = -\int E_1^i \partial_j E_2^j \, d^3x = -\frac{q_2}{\epsilon_0} E_1^i(\mathbf{r}_2) = -\frac{q_1 q_2}{4\pi\epsilon_0^2} \frac{n^i}{d^2}.$$

Evaluate the same derivative on the parametrised form, using $\partial_j(1/d) = -n^j/d^2$ and $\partial_j(d^i d^j/d^3) = n^i/d^2$:

$$\frac{\partial}{\partial d^j} \left[\frac{A\delta^{ij} + Bn^i n^j}{d} \right] = -\frac{A n^i}{d^2} + \frac{B n^i}{d^2} = \frac{(B - A) n^i}{d^2}.$$

Comparing, $B - A = -q_1 q_2 / 4\pi\epsilon_0^2$. Solve the two equations: $A = q_1 q_2 / 8\pi\epsilon_0^2$ and $B = -A$. Hence

$$M^{ij} = \frac{q_1 q_2}{8\pi\epsilon_0^2 d} (\delta^{ij} - n^i n^j),$$

which is (2.6.98). Note the structure: $\delta - nn$ is the projector onto directions *perpendicular* to the line joining the charges, so $M^{ij} n^j = 0$ — the longitudinal part of the integral vanishes identically. This

was also checked by direct numerical integration before being written down.

▼ Grind box — differentiating (2.6.99)

Write $C = \mu_0 q_1 q_2 / 8\pi$ and $\mathbf{V} = v(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, a constant. Then

$$\mathbf{P}_{\text{field}} = C \left[\frac{\mathbf{V}}{d} + \frac{\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{V})}{d} \right].$$

The kinematics at $t = 0$. $\mathbf{d}(t) = \mathbf{r}_2 - \mathbf{r}_1 = (d - vt)\hat{\mathbf{x}} + vt\hat{\mathbf{y}}$, so $\dot{\mathbf{d}} = v(\hat{\mathbf{y}} - \hat{\mathbf{x}})$ and

$$\dot{d} = \hat{\mathbf{n}} \cdot \dot{\mathbf{d}} = \hat{\mathbf{x}} \cdot v(\hat{\mathbf{y}} - \hat{\mathbf{x}}) = -v, \quad \dot{\hat{\mathbf{n}}} = \frac{\dot{\mathbf{d}}}{d} - \frac{\mathbf{d} \dot{d}}{d^2} = \frac{v(\hat{\mathbf{y}} - \hat{\mathbf{x}})}{d} + \frac{v\hat{\mathbf{x}}}{d} = \frac{v}{d}\hat{\mathbf{y}}.$$

Note $\dot{\hat{\mathbf{n}}} \perp \hat{\mathbf{n}}$, as it must be for a unit vector.

First term.

$$\frac{d}{dt} \frac{\mathbf{V}}{d} = -\frac{\dot{d}}{d^2} \mathbf{V} = \frac{v}{d^2} v(\hat{\mathbf{x}} + \hat{\mathbf{y}}) = \frac{v^2}{d^2} (\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

Second term. Three pieces by the product rule:

$$\begin{aligned} \frac{d}{dt} \left[\frac{\hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{V})}{d} \right] &= -\frac{\dot{d}}{d^2} \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{V}) + \frac{\dot{\hat{\mathbf{n}}}(\hat{\mathbf{n}} \cdot \mathbf{V})}{d} + \frac{\hat{\mathbf{n}}(\dot{\hat{\mathbf{n}}} \cdot \mathbf{V})}{d} \\ &= \frac{v}{d^2} \hat{\mathbf{x}}(v) + \frac{(v/d)\hat{\mathbf{y}}(v)}{d} + \frac{\hat{\mathbf{x}}((v/d)\hat{\mathbf{y}} \cdot v(\hat{\mathbf{x}} + \hat{\mathbf{y}}))}{d} \\ &= \frac{v^2}{d^2} \hat{\mathbf{x}} + \frac{v^2}{d^2} \hat{\mathbf{y}} + \frac{v^2}{d^2} \hat{\mathbf{x}} = \frac{v^2}{d^2} (2\hat{\mathbf{x}} + \hat{\mathbf{y}}). \end{aligned}$$

Total.

$$\frac{d\mathbf{P}_{\text{field}}}{dt} = C \frac{v^2}{d^2} \left[(\hat{\mathbf{x}} + \hat{\mathbf{y}}) + (2\hat{\mathbf{x}} + \hat{\mathbf{y}}) \right] = \frac{\mu_0 q_1 q_2 v^2}{8\pi d^2} (3\hat{\mathbf{x}} + 2\hat{\mathbf{y}}),$$

which is (2.6.101) after pulling out a factor of 2: $\frac{1}{8\pi} (3\hat{\mathbf{x}} + 2\hat{\mathbf{y}}) = \frac{1}{4\pi} \left(\frac{3}{2}\hat{\mathbf{x}} + \hat{\mathbf{y}} \right)$. ✓ Every term of this box, and of (2.6.94), was verified symbolically; the sum (2.6.102) is zero identically, not numerically.

WHERE T^{MN} GOES NEXT

You have built a symmetric, conserved, traceless rank-2 tensor whose 00 component is energy density and whose $0i$ components are momentum density. Chapter 3.6 will take Einstein's field equation

$$G^{\mu\nu} = \frac{8\pi G}{c^4} T^{\mu\nu}$$

and put *this object* on the right-hand side. That is what "energy gravitates" means technically: not mass but $T^{\mu\nu}$ sources spacetime curvature, so the electromagnetic field itself — light, a magnetic field, the energy in a capacitor — gravitates, with a strength you can now compute. It is also why the equation had to be symmetric in $\mu\nu$, which is why §10.1's improvement term was not optional. And the tracelessness $\eta_{\mu\nu} T^{\mu\nu} = 0$, which you can verify in one line from (2.6.80), is the statement that the photon is massless, and reappears in Chapter 5.9 as the conformal symmetry of classical electromagnetism — a symmetry that quantum corrections break.

11 · Worked examples

WORKED EXAMPLE 1 — THE WIRE, IN FULL NUMBERS

A copper wire of cross-section 1 mm^2 carries 1 A . A proton ($q = +e$) travels parallel to it at $u = 10^5 \text{ m s}^{-1}$, at a perpendicular distance $r = 1 \text{ cm}$. Compute the force on it twice: magnetically in the lab, and electrostatically in its own rest frame. Then say how big the charge imbalance is.

The lab. The field of the wire at 1 cm is

$$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7})(1)}{2\pi(0.01)} = 2.000 \times 10^{-5} \text{ T},$$

about a third of the Earth's field. The magnetic force is

$$F_{\text{lab}} = quB = (1.602 \times 10^{-19})(10^5)(2.000 \times 10^{-5}) = 3.204 \times 10^{-19} \text{ N},$$

directed toward the wire. No electric force: the wire is neutral in this frame.

The wire's insides. From (2.6.49), $n = 8.49 \times 10^{28} \text{ m}^{-3}$, so the mobile charge per metre is

$$\lambda_0 = neA = (8.49 \times 10^{28})(1.602 \times 10^{-19})(10^{-6}) = 1.360 \times 10^4 \text{ C m}^{-1},$$

and the drift speed is $v_d = I/\lambda_0 = 7.351 \times 10^{-5} \text{ m s}^{-1}$, i.e. 0.0735 mm s^{-1} .

The proton's frame. Here $u/c = 3.34 \times 10^{-4}$, so $\gamma_u = 1.000000056$ — utterly negligible, but we keep it because the whole effect is of this size. From (2.6.45),

$$\begin{aligned} \lambda' &= -\gamma_u \lambda_0 \frac{uv_d}{c^2} = -\frac{(1.360 \times 10^4)(10^5)(7.351 \times 10^{-5})}{8.988 \times 10^{16}} \\ &= -1.113 \times 10^{-12} \text{ C m}^{-1}. \end{aligned}$$

The electric field of that line charge at $r = 1 \text{ cm}$ is

$$E' = \frac{\lambda'}{2\pi\epsilon_0 r} = \frac{-1.113 \times 10^{-12}}{2\pi(8.854 \times 10^{-12})(0.01)} = -2.000 \text{ V m}^{-1},$$

pointing toward the wire, and the force on the proton is

$$F' = qE' = (1.602 \times 10^{-19})(2.000) = 3.204 \times 10^{-19} \text{ N}.$$

Compare. $F'/F_{\text{lab}} = 1.000000056 = \gamma_u \checkmark$, exactly as (2.6.47) requires. Notice also the clean intermediate result $E' = -uB$ to this accuracy — which is just $E'_y = \gamma(E_y - uB_z)$ with $E_y = 0$, the field transformation rule of §5, arriving by a completely independent route.

How big is the imbalance? In electrons per metre:

$$\frac{|\lambda'|}{e} = 6.94 \times 10^6 \text{ m}^{-1}, \quad \text{against} \quad \frac{\lambda_0}{e} = 8.49 \times 10^{22} \text{ m}^{-1}.$$

A fractional imbalance of 8.2×10^{-17} — one part in twelve million billion. That is the entire magnetic force on this proton: a discrepancy in the seventeenth significant figure of a cancellation. If the positive and negative charge in a wire did not cancel to *far* better than that, the residual electrostatic force would swamp every magnetic effect ever measured, and nobody would have discovered magnetism in a laboratory at all.

WORKED EXAMPLE 2 — A BOOST THAT ELIMINATES THE ELECTRIC FIELD

In the lab, $\mathbf{E} = E_0 \hat{\mathbf{y}}$ and $\mathbf{B} = B_0 \hat{\mathbf{z}}$ are uniform, crossed, and satisfy $E_0 < cB_0$. Find a frame in which the electric field vanishes, describe the motion of a charge in that frame, and identify the frame velocity in a form that does not refer to the axes.

Check first that it is possible. The invariants of §7 are $B_0^2 - E_0^2/c^2 > 0$ and $\mathbf{E} \cdot \mathbf{B} = 0$, which is precisely condition (2.6.62). So a frame with $\mathbf{E}' = \mathbf{0}$ is permitted. Now construct it.

The construction. Boost along $\hat{\mathbf{x}}$ at speed v . From §5, $E'_x = E_x = 0$ and $E'_z = \gamma(E_z + vB_y) = 0$ automatically. The only surviving component is

$$E'_y = \gamma(E_0 - vB_0),$$

and since γ is never zero, $\mathbf{E}' = \mathbf{0}$ requires

$$v = \frac{E_0}{B_0}.$$

This is subluminal precisely when $E_0 < cB_0$ — the invariant condition, arriving as a kinematic constraint rather than as a separate assumption. Note in passing what happens if $E_0 > cB_0$: the required v exceeds c , there is no such frame, and the invariant told you so in advance without any construction.

Axis-free form. With $\mathbf{E} \times \mathbf{B} = E_0 B_0 (\hat{\mathbf{y}} \times \hat{\mathbf{z}}) = E_0 B_0 \hat{\mathbf{x}}$ and $B^2 = B_0^2$,

$$\frac{\mathbf{E} \times \mathbf{B}}{B^2} = \frac{E_0}{B_0} \hat{\mathbf{x}} = \mathbf{v}.$$

So the frame in question moves with the **drift velocity**

$$\mathbf{v}_{\text{drift}} = \frac{\mathbf{E} \times \mathbf{B}}{B^2}$$

— a formula with no reference to any choice of axes, valid whenever $\mathbf{E} \perp \mathbf{B}$ and $E < cB$.

What the motion looks like. In the primed frame there is only a magnetic field, of magnitude

$$B' = \gamma \left(B_0 - \frac{v}{c^2} E_0 \right) = \gamma B_0 \left(1 - \frac{E_0^2}{B_0^2 c^2} \right) = \frac{B_0}{\gamma} = \sqrt{B_0^2 - \frac{E_0^2}{c^2}},$$

using $v = E_0/B_0$ and $\gamma^2(1 - v^2/c^2) = 1$. The last expression is the square root of the first invariant divided by two — as it must be, since with $\mathbf{E}' = \mathbf{0}$ the invariant (2.6.56) reads $2B'^2$, and it has to agree with $2(B_0^2 - E_0^2/c^2)$. A charge in a pure magnetic field feels a force always perpendicular to its velocity, does no work ((2.6.69)), and therefore moves in a **circle** at constant speed, the cyclotron gyration.

Back in the lab, that circle is being carried along at $\mathbf{v}_{\text{drift}}$: the trajectory is a cycloid-like drift, and — this is the point — *the drift velocity does not depend on the charge, the sign of the charge, or the mass*. Every particle drifts at $\mathbf{E} \times \mathbf{B}/B^2$.

Why every mass spectrometer contains one. Turn the argument around. A particle that enters crossed fields moving at exactly v_{drift} is, in the primed frame, at rest — and a charge at rest in a pure magnetic field feels no force at all. So it passes through *undeflected*, while anything faster or slower is bent aside. That is a **velocity selector**: a slit, two plates, a magnet, and one equation. It selects $v = E/B$ regardless of mass and charge, which is exactly what you want in front of a mass analyser that will then separate by m/q . The same $\mathbf{E} \times \mathbf{B}$ drift governs charged particles in the magnetosphere, the confinement of tokamak plasmas, and the Hall effect.

12 · Your turn

PROBLEM 1 — THE FIELD OF A CHARGE IN UNIFORM MOTION

A charge q moves with constant velocity $v\hat{\mathbf{x}}$, passing the origin at $t = 0$. By boosting the Coulomb field from its rest frame, show that at $t = 0$ the lab field at position $\mathbf{r} = (x, y, z)$ is

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3} \frac{1 - \beta^2}{(1 - \beta^2 \sin^2 \theta)^{3/2}}, \quad \mathbf{B} = \frac{1}{c^2} \mathbf{v} \times \mathbf{E},$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{v}$. Show the field is still *radial* from the present position, that it is weakened by γ^2 directly ahead and strengthened by γ broadside, and that for $\beta \ll 1$ the magnetic field reduces to the Biot-Savart expression Chapter 1.1 had to quote. Comment on what the field looks like as $\gamma \rightarrow \infty$.

– Solution

Set-up. Let S' be the charge's rest frame, moving at $+v\hat{\mathbf{x}}$. There $\mathbf{B}' = \mathbf{0}$ and $\mathbf{E}' = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}'}{r'^3}$.

Coordinates. The lab event is $(ct, x, y, z) = (0, x, y, z)$. By the Lorentz transformation, $x' = \gamma(x - vt) = \gamma x$, $y' = y$, $z' = z$, so

$$r' = \sqrt{\gamma^2 x^2 + y^2 + z^2}.$$

Fields. Transform back from S' to the lab. Parallel components are unchanged and perpendicular ones acquire a γ (with $\mathbf{B}' = \mathbf{0}$ the $\mathbf{v} \times \mathbf{B}$ terms drop):

$$E_x = E'_x = \frac{q}{4\pi\epsilon_0} \frac{\gamma x}{r'^3}, \quad E_y = \gamma E'_y = \frac{q}{4\pi\epsilon_0} \frac{\gamma y}{r'^3}, \quad E_z = \frac{q}{4\pi\epsilon_0} \frac{\gamma z}{r'^3}.$$

All three carry the same factor γ/r'^3 , so

$$\mathbf{E} = \frac{q\gamma}{4\pi\epsilon_0} \frac{\mathbf{r}}{r'^3}.$$

The field points radially away from the charge's present position — a small miracle, given that information travels at c and the charge has moved since the field "left". It works only for *uniform* motion; accelerate the charge and the radial structure breaks, which is what radiation is.

The angular factor. Write $x = r \cos \theta$ and $y^2 + z^2 = r^2 \sin^2 \theta$:

$$r'^2 = \gamma^2 r^2 \cos^2 \theta + r^2 \sin^2 \theta = \gamma^2 r^2 \left[\cos^2 \theta + \frac{\sin^2 \theta}{\gamma^2} \right] = \gamma^2 r^2 [1 - \beta^2 \sin^2 \theta],$$

using $1/\gamma^2 = 1 - \beta^2$ and $\cos^2 + \sin^2 = 1$. Then $r'^3 = \gamma^3 r^3 (1 - \beta^2 \sin^2 \theta)^{3/2}$ and

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3} \cdot \frac{\gamma}{\gamma^3 (1 - \beta^2 \sin^2 \theta)^{3/2}} = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3} \cdot \frac{1 - \beta^2}{(1 - \beta^2 \sin^2 \theta)^{3/2}}. \quad \checkmark$$

The two special directions. Ahead or behind ($\theta = 0, \pi$): the factor is $1 - \beta^2 = 1/\gamma^2$, so the field is *weaker* than Coulomb by γ^2 . Broadside ($\theta = \pi/2$): the factor is $(1 - \beta^2)/(1 - \beta^2)^{3/2} = (1 - \beta^2)^{-1/2} = \gamma$, so the field is *stronger* by γ . These are exactly (2.6.92) and (2.6.93), which §10 needed and derived by the same argument in two special cases.

The magnetic field. Transforming $\mathbf{B}$ from a frame where it vanishes gives $B_y = -\gamma v E'_z/c^2$ and $B_z = +\gamma v E'_y/c^2$, which in view of the expressions above is exactly $\mathbf{B} = \mathbf{v} \times \mathbf{E}/c^2$ — *exactly*, not to leading order. For $\beta \ll 1$ we may replace $\mathbf{E}$ by its Coulomb value:

$$\mathbf{B} \approx \frac{1}{c^2} \mathbf{v} \times \frac{q\hat{\mathbf{r}}}{4\pi\epsilon_0 r^2} = \frac{\mu_0}{4\pi} \frac{q \mathbf{v} \times \hat{\mathbf{r}}}{r^2},$$

using $1/(\epsilon_0 c^2) = \mu_0$. That is the Biot–Savart law for a point charge — the second of the two results Chapter 1.1 §3.3 had to quote, now derived. Both of that chapter's ■ items are discharged.

The pancake. As $\gamma \rightarrow \infty$ the forward field dies like $1/\gamma^2$ while the broadside field grows like γ . The angular width over which the field is appreciable is set by $\beta^2 \sin^2 \theta \approx 1$, i.e. $\sin \theta \approx 1/\gamma$: the field collapses into a disc perpendicular to the motion, of angular thickness $\sim 1/\gamma$, travelling with the charge. At the LHC, $\gamma \approx 7000$ for protons, so each beam's field is squashed into a pancake about 10^{-4} radians thick. That is why an ultrarelativistic charged particle acts on a target like a *pulse* rather than a slowly growing force, and it is the starting point for the equivalent-photon approximation used to describe ultraperipheral collisions in Part V.

PROBLEM 2 — A PLANE WAVE HAS NOTHING TO GIVE UP

For a plane electromagnetic wave in vacuum, $\mathbf{E} \perp \mathbf{B}$ and $E = cB$. Show that both invariants of §7 vanish. Then use that to prove three things: no boost can eliminate either field; no boost can make the fields non-perpendicular; and no boost can bring the wave to rest. What *can* a boost do to a light wave?

– Solution

The invariants. $F_{\mu\nu}F^{\mu\nu} = 2(B^2 - E^2/c^2) = 2(B^2 - B^2) = 0$, and $\epsilon_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma} = (8/c)\mathbf{E} \cdot \mathbf{B} = 0$ since the fields are perpendicular. Both are zero. By the invariance theorem they are zero in *every* frame.

(i) Neither field can be eliminated. Suppose some frame had $\mathbf{B}' = \mathbf{0}$. Then in that frame $F_{\mu\nu}F^{\mu\nu} = -2E'^2/c^2$, which must equal zero, forcing $E' = 0$ as well. So the only way to lose the magnetic field is to lose the whole wave — and you cannot, since a tensor that vanishes in one frame vanishes in all (Chapter 2.4 §6), so a wave that exists at all exists for everyone. Identically for $\mathbf{E}' = \mathbf{0}$.

(ii) The fields stay perpendicular and stay in ratio. $\mathbf{E}' \cdot \mathbf{B}' = 0$ in every frame because the pseudoscalar is invariant, and $E' = cB'$ in every frame because the scalar is. A light wave looks like a light wave to everybody.

(iii) It cannot be brought to rest. "At rest" would mean a static field configuration, for which $\mathbf{E}$ and $\mathbf{B}$ are independent and generically $E \neq cB$. More sharply: the wave's four-wavevector $k^\mu = (\omega/c, \mathbf{k})$ satisfies $k \cdot k = 0$ (Chapter 2.5 §7.1), and a null four-vector cannot be boosted to a purely timelike one — that is the same statement as " $\Delta s^2 = 0$ is preserved" in Chapter 2.3. The vanishing of both field invariants is the field-theoretic face of the same fact: null is a Lorentz-invariant category.

What a boost *can* do. Change the amplitude and the frequency, together and in the same ratio. From §5, boosting along the propagation direction multiplies both $\mathbf{E}_\perp$ and $\mathbf{B}_\perp$ by the Doppler factor $\sqrt{(1 - \beta)/(1 + \beta)}$, exactly the factor Chapter 2.5 §7.2 derived for ω . So a light wave can be made as weak and as red as you like — asymptotically, but never exactly, zero. It can also be aimed differently: a boost transverse to $\mathbf{k}$ changes the propagation direction, which is aberration. What survives every boost is the wave's *identity* as a wave.

The physical moral. The vanishing of both invariants is what makes light structurally different from every static field. It is also why the photon is massless: the null condition on k^μ is $E = pc$ (Chapter 2.5 §4.3), and $m = 0$.

PROBLEM 3 — CHARGE CONSERVATION AS AN OUTPUT, NOT AN INPUT

Show that $\partial_\nu \partial_\mu F^{\mu\nu} = 0$ identically, for *any* antisymmetric $F^{\mu\nu}$. Deduce that $\partial_\nu j^\nu = 0$ follows from the field equation rather than being an extra assumption. Then explain what this says about a hypothetical universe in which charge is not conserved, and connect it to Chapter 0.7's Problem 4 on the displacement current.

– Solution

The identity. The object $\partial_\nu \partial_\mu$ is symmetric under $\mu \leftrightarrow \nu$ (mixed partials commute) and $F^{\mu\nu}$ is antisymmetric. Contracting a symmetric object with an antisymmetric one over both indices gives zero — the argument of (2.6.71), repeated. Explicitly, relabel the two dummy indices:

$$X \equiv \partial_\nu \partial_\mu F^{\mu\nu} = \partial_\mu \partial_\nu F^{\nu\mu} = -\partial_\nu \partial_\mu F^{\mu\nu} = -X \implies X = 0.$$

The consequence. Take ∂_ν of the field equation (2.6.19):

$$0 = \partial_\nu \partial_\mu F^{\mu\nu} = \mu_0 \partial_\nu j^\nu \implies \partial_\nu j^\nu = 0.$$

Charge conservation is a *theorem*, not a postulate. Section 1 assumed it in order to build j^μ ; the theory then hands it back.

What it forbids. This is stronger than it looks. It says you cannot write down a consistent electrodynamics with a non-conserved source. If someone hands you a j^μ with $\partial_\nu j^\nu \neq 0$, the equation $\partial_\mu F^{\mu\nu} = \mu_0 j^\nu$ has *no solutions at all* — not "solutions with strange behaviour", none. A universe in which charge could appear from nowhere could not have Maxwell's equations, in any frame, even approximately. The rigidity comes entirely from the antisymmetry of F , which came from gauge invariance (the callout at the end of §2), which is why Chapter 6.3 can say that gauge symmetry and charge conservation are two descriptions of one fact.

The displacement current. Chapter 0.7's Problem 4 asked you to show that $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ is *inconsistent* with charge conservation, because taking the divergence gives $0 = \mu_0 \nabla \cdot \mathbf{J}$, which contradicts (2.6.1) whenever $\partial_t \rho \neq 0$ — and that the unique minimal repair is Maxwell's $\mu_0 \epsilon_0 \partial_t \mathbf{E}$. This problem is the four-dimensional version of that argument, and it is now visibly the *same* argument: the three-dimensional identity $\nabla \cdot (\nabla \times \mathbf{B}) = 0$ that forced the repair is the spatial part of $\partial_\nu \partial_\mu F^{\mu\nu} = 0$. In the covariant formulation the displacement current is not a repair. It was never absent.

PROBLEM 4 — TWO PARALLEL WIRES, BOTH WAYS

Two long parallel wires a distance d apart carry currents I_1 and I_2 in the same direction. Model each as in §6: lattice $+\lambda_0$ at rest, electrons $-\lambda_0$ drifting at $-v_a\hat{\mathbf{x}}$, so that $I_a = \lambda_0 v_a$. **(a)** Compute the force per unit length magnetically, in the lab. **(b)** Now boost to the rest frame of wire 2's conduction electrons and recompute it electrostatically, being careful about how "per unit length" transforms. **(c)** Check that wire 2's *lattice* feels no net force in that frame either, as it must not. **(d)** Put in $I_1 = I_2 = 1$ A and $d = 1$ cm.

– Solution

(a) The lab. Wire 1 produces $\mathbf{B}_1 = \frac{\mu_0 I_1}{2\pi d} \hat{\mathbf{z}}$ at wire 2 (taking wire 2 to lie in the $+\hat{\mathbf{y}}$ direction from wire 1). Only wire 2's electrons move, with charge per unit length $-\lambda_0$ and velocity $-v_2\hat{\mathbf{x}}$:

$$\frac{\mathbf{f}}{L} = (-\lambda_0)(-v_2\hat{\mathbf{x}}) \times \frac{\mu_0 I_1}{2\pi d} \hat{\mathbf{z}} = \lambda_0 v_2 \frac{\mu_0 I_1}{2\pi d} (\hat{\mathbf{x}} \times \hat{\mathbf{z}}) = -\frac{\mu_0 I_1 I_2}{2\pi d} \hat{\mathbf{y}}.$$

Attractive, magnitude $\mu_0 I_1 I_2 / 2\pi d$. The lattice of wire 2 is at rest and wire 1 is neutral, so nothing else contributes.

(b) The co-moving frame. Boost at $w = -v_2$ (so wire 2's electrons are at rest). By (2.6.45) with $u \rightarrow w$, wire 1 acquires

$$\lambda'_1 = -\gamma_2 \lambda_0 \frac{wv_1}{c^2} = +\gamma_2 \frac{\lambda_0 v_1 v_2}{c^2} = +\gamma_2 \frac{I_1 v_2}{c^2}, \quad \gamma_2 = \left(1 - \frac{v_2^2}{c^2}\right)^{-1/2},$$

a *positive* charge density — and wire 2's electrons are negative, so the force is attractive, agreeing with (a) in direction.

Wire 2's electrons are now at rest, so their density is their proper density, $-\lambda_0/\gamma_2$ (they were contracted by γ_2 in the lab). The electric field from wire 1 at distance d is $E'_y = \lambda'_1 / 2\pi\epsilon_0 d$, so the force per unit length *measured in this frame* is

$$\frac{f'_y}{L'} = \left(-\frac{\lambda_0}{\gamma_2}\right) \frac{\gamma_2 \lambda_0 v_1 v_2 / c^2}{2\pi\epsilon_0 d} = -\frac{\lambda_0^2 v_1 v_2}{2\pi\epsilon_0 c^2 d} = -\frac{\mu_0 I_1 I_2}{2\pi d}.$$

The "per unit length" bookkeeping. That looks like the same number as (a), which would be wrong, because transverse forces are *not* equal in the two frames — they differ by γ ((2.6.48)). The resolution is that lengths differ too. Take a lab segment of length L containing N electrons. Those same electrons are at rest in the primed frame, so they occupy $L' = \gamma_2 L$ there. Hence the *total* force on them is

$$F'_y = \frac{f'_y}{L'} \cdot L' = -\frac{\mu_0 I_1 I_2}{2\pi d} \gamma_2 L \quad \implies \quad F_y = \frac{F'_y}{\gamma_2} = -\frac{\mu_0 I_1 I_2}{2\pi d} L. \quad \checkmark$$

The two factors of γ_2 — one from length contraction, one from force transformation — cancel, which is exactly why the force *per unit length* happens to come out numerically the same in both frames. That

coincidence is worth noticing precisely so that you do not mistake it for a proof; the total force on a given set of charges is the frame-covariant statement.

(c) The lattice of wire 2. In the primed frame it moves at $+v_2\hat{\mathbf{x}}$, has density $\gamma_2\lambda_0$, and sits in both an electric and a magnetic field. From §5, $B'_z = \gamma_2(B_z - wE_y/c^2) = \gamma_2\mu_0 I_1/2\pi d$ since $E_y = 0$ in the lab. So

$$\begin{aligned} \text{electric: } & (\gamma_2\lambda_0) \frac{\gamma_2\lambda_0 v_1 v_2 / c^2}{2\pi\epsilon_0 d} \hat{\mathbf{y}} = +\gamma_2^2 \frac{\mu_0 I_1 I_2}{2\pi d} \hat{\mathbf{y}}, \\ \text{magnetic: } & (\gamma_2\lambda_0)(v_2\hat{\mathbf{x}}) \times \left(\gamma_2 \frac{\mu_0 I_1}{2\pi d} \hat{\mathbf{z}} \right) = -\gamma_2^2 \frac{\mu_0 I_1 I_2}{2\pi d} \hat{\mathbf{y}}. \end{aligned}$$

They cancel exactly. The lattice feels nothing, as it must, since in the lab it feels nothing and "zero force" is a frame-independent statement. Notice how the cancellation works: in this frame the lattice is a current, and the repulsion from the now-charged wire 1 is precisely balanced by the magnetic attraction between the two currents. Neither description is more true.

(d) Numbers.

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(4\pi \times 10^{-7})(1)(1)}{2\pi(0.01)} = 2 \times 10^{-5} \text{ N m}^{-1}.$$

Twenty micronewtons per metre — small, but measurable with a torsion balance, and until 2019 this configuration *defined* the ampere: the current which, in two infinite parallel wires one metre apart, produces a force of $2 \times 10^{-7} \text{ N m}^{-1}$. An entire base unit of the SI was defined by a relativistic correction to Coulomb's law of relative order 10^{-26} .

Six objects, and Part II is finished.

The four-current $j^\mu = (c\rho, \mathbf{J})$, built by requiring $\partial_\mu j^\mu = 0$ to reproduce Chapter 0.7's continuity equation — after which charge conservation is one contracted index and therefore true in every frame. **The field tensor** $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$, whose six independent components are $-\mathbf{E}/c$ in the first row and $-\epsilon^{ijk} B^k$ in the spatial block — the six that Chapter 2.4 §7.2 counted and promised.

Maxwell's four equations as two: $\partial_\mu F^{\mu\nu} = \mu_0 j^\nu$ expanding to Gauss and Ampère–Maxwell, and $\partial_{[\lambda} F_{\mu\nu]} = 0$ expanding to $\nabla \cdot \mathbf{B} = 0$ and Faraday — the second of which is not physics at all but six terms cancelling by Clairaut, which is Chapter 0.7's $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ and $\nabla \times \nabla\phi = \mathbf{0}$ fused into one identity. **Gauge freedom** $A^\mu \rightarrow A^\mu + \partial^\mu \chi$, Chapter 1.2's total-derivative freedom in field form, spent on the Lorenz gauge to give $\square A^\mu = \mu_0 j^\mu$. **Two invariants**, $2(B^2 - E^2/c^2)$ and $(8/c)\mathbf{E} \cdot \mathbf{B}$, which classify field configurations the way the sign of Δs^2 classifies intervals. **The Lagrangian** $-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - j_\mu A^\mu$, entry three in Chapter 1.2's table, which on variation returns the field equations and which Chapter 6.4 will copy verbatim with a non-abelian F . And **the stress-energy tensor**, whose T^{00} is $\frac{1}{2}(\epsilon_0 E^2 + B^2/\mu_0)$ and whose T^{0i} is $\mathbf{S}/c$, giving field momentum density $\mathbf{g} = \epsilon_0 \mathbf{E} \times \mathbf{B}$.

And two long-standing debts, both settled by name. Chapter 2.1 posed a contradiction between Maxwell and Galileo and offered a fork; the resolution is that Maxwell's equations single out a *speed*, which lives in the metric, and never singled out a *frame* — so branch (A) was not merely unsupported by Michelson and Morley, it was structurally impossible. Chapter 1.1 showed Newton's third law failing for two moving charges by $-\mu_0 q_1 q_2 v^2 \hat{\mathbf{y}}/4\pi d^2$ and wrote an IOU; §10 found the missing momentum sitting in the field at $\mu_0 q_1 q_2 v(2\hat{\mathbf{x}} + \hat{\mathbf{y}})/8\pi d$, showed its rate of change cancels the mechanical one exactly — in both components, including a $\frac{3}{2}$ that 1.1 never saw — and thereby forced you to accept that a field is not a bookkeeping device but a physical system.

And the thesis, which is the reason the chapter exists. Electromagnetism did not need to be made compatible with relativity. It was relativistic before the word existed, which is exactly why it produced a frame-independent c and broke the physics of 1900. What needed fixing was mechanics, and Chapters 2.2 to 2.5 fixed it. The magnetic field is not a second force alongside the electric one; it is what an electric field looks like from a moving frame — an effect of relative order $u v_d/c^2$, which for two ordinary currents in copper is about 10^{-26} , and which is visible at all only because the positive and negative charge in matter cancel to *at least* that precision. A discrepancy in the twenty-sixth significant figure, and it defines a base unit of the SI. Two fields, one tensor; four equations, two lines; and a c that belongs to spacetime rather than to any medium. That is the whole of Part II in one chapter, and it is the last thing you need before geometry starts to move.

Where this gets spent.

- $T^{\mu\nu}$ — **Chapter 3.6:** it is what sits on the right-hand side of the Einstein field equations. Energy gravitates, and now you know what "energy" means as a tensor and why it had to be symmetric.

- $F = dA$ **and** $\partial_{[\lambda} F_{\mu\nu]} = 0$ → Chapter 3.5, where they become $F = dA$ and $dF = 0$ literally, and $d^2 = 0$ stops being an analogy.
- $\mathcal{L} = -\frac{1}{4\mu_0} F^2 - j \cdot A$ → Chapter 5.2, quantised, where the field's normal modes become photons; Chapter 5.6, where $-j_\mu A^\mu$ becomes the single vertex of QED and every Feynman diagram in the theory is built from copies of it.
- **Gauge freedom** → Chapter 6.3, promoted from a convenience to the *generating principle*: demand it locally and this entire chapter is the output rather than the input. Then Chapter 6.4 repeats the construction with a non-commuting group and gets Yang–Mills, using §9's Lagrangian unchanged except for a commutator.
- **Charge invariance and the E–B mixing** → Chapter 5.4, where the Dirac field's coupling to A^μ is fixed by exactly these transformation properties.
- **The whole structure** → Chapter 7.1, where the gauge fields of the Standard Model become excitations of a string, and $F_{\mu\nu} F^{\mu\nu}$ is the leading term of an expansion whose higher terms are the Born–Infeld action.